PHYSICS MODULO GAUGE

The Methodology of the Incompleteness Framework

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ABSTRACT

The Incompleteness framework proves an exact finite observable-law equivalence with fixed-basis unitary/Born representation, identifies the memory-bearing sector through embedded-observer structure, and isolates the common coherent local instrument/composite lift as the remaining operational boundary — and derives, conditionally on the cubic-lattice commitment and the physical-carrier bridge H-link ([SM §4.5]), the Standard Model gauge structure, together with a route to general relativity, from a single starting point — that observation occurs — together with the structural conditions that an embedded observer must satisfy. This paper articulates the framework's *methodology*, separately from its physics, for readers in the foundations and philosophy-of-physics community. Five interrelated commitments organize the account. First, the framework's foundation, examined directly, resolves into a *two-axiom observation base*: an indubitable axiom (tokened differentiation occurs) and a substantive second axiom (differentiation recurs), the second presupposing but not derivable from the first — a no-go result we state and defend. Second, the framework realizes a scoped *constructive structural realism*: the **local propagating lattice/gauge residue** consistent with the empirical inputs is uniquely fixed under the reconstruction theorem's stated conditions, while Bell-inclusive existence is conditional on H-Bell and Bell-inclusive uniqueness remains open; the residual gauge freedom of the reconstructed dilation/local datum is explicitly exhibited, with the completeness of its classification open. Third, the framework's substratum, reduced description and trace-out provide a common representation in which to state the distinct premises of Bell, Kochen–Specker, Pusey–Barrett–Rudolph and Frauchiger–Renner. The premises require separate checks: the adopted Bell branch requires ontic parameter dependence and H-Bell, while the other cases retain their stated operational-algebra, preparation-independence and observer-consistency assumptions. Fourth, these combine into a general methodological proposal — that the well-posed question for fundamental physics is not "what is fundamental?" but "what is the equivalence class of structures producing observations, and what are its representatives?" — *physics modulo gauge*. Fifth, Part V develops the operational form of the methodology: a gauge-invariance no-go on substrate-only computation and a category-error finding on substrate-level Lagrangians sharpen the boundary between substrate-derivable content and content selected by empirical input; the two results are compared with Part III's distinct premise checks using the same substratum and observable descriptions; systematic audits applying the operational standard to the framework's

own content are summarized; and three methodological refinements canonized by the audit experience are stated at the generality the methodology requires. The paper develops each commitment and argues that, taken together, they constitute a replicable methodology rather than features peculiar to one framework. Developmental status is made explicit throughout: the foundational distillation of Part I is recent and is offered for scrutiny; the structural results of Part II are summarized from the framework's technical papers at the fidelity of faithful synthesis; Part III's four no-go cases divide by status — two (Bell, Frauchiger–Renner) developed results summarized from the framework's technical papers, two (Kochen–Specker, Pusey–Barrett–Rudolph) developed in this paper as *conditional* chains on results those papers already establish, with the noncommuting algebra and tensor composite taken as given by the operational-completion characterization and the preparation and observer-consistency premises checked separately ([Main] §3.2, §3.4); and Part V is more recent still — its two no-go results are developed in the framework's technical record, the systematic audits it summarizes are developed there as well, and the comparison with Part III and the three operational refinements constitute a methodological proposal, with the substantive premises of each result stated separately.

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1. INTRODUCTION

1.1 What this paper is, and what it is not

This is a paper about method. The Incompleteness framework is a body of physics — it makes quantitative, parameter-free retrodictions across the Standard Model, conditional on the cubic-lattice commitment and the bridge hypotheses [SM] names (H-link for the gauge structure, [SM §4.5]) (including a value matching the post-JUNO global fit at 0.07σ and the direct JUNO measurement at 0.14σ) and recovers the Bekenstein–Hawking entropy coefficient as a consistency relation of its horizon calibration — and that physics is developed in a sequence of technical papers — [Main], [SM], [Substratum], [GR] — to which this one is a companion. The present paper does not re-derive any of it. It asks a different question: *what is the framework's methodology*, and is that methodology defensible and replicable independently of whether the framework's particular physical claims survive?

The question is worth separating because the framework's methodological posture is unusual in ways that a reader assessing the physics might not pause over, but that a reader in foundations or philosophy of physics will recognize as substantive. The framework claims a single foundational commitment and derives the rest. It claims that what other programs treat as free assumptions are, in its setting, forced. It claims a determinate relationship to the major no-go theorems of quantum foundations. And it claims that a long-standing question in the metaphysics of physics — the realism debate — is, as usually posed, the wrong question. Each of these is a methodological claim, evaluable

on its own terms, and each is the subject of one part of this paper.

What this paper is not: it is not a defense of the framework's physics, which must stand on the technical papers and their empirical record; it is not a survey of the foundations literature, though it engages that literature where engagement is due; and it is not a finished statement. Part I in particular presents a recent distillation of the framework's foundational commitment, offered for scrutiny rather than asserted as settled. We mark developmental status explicitly wherever it applies.

1.2 The five commitments

The paper is organized around five methodological commitments, developed in Parts I through V.

Part I — the foundational commitment. The framework is often described as resting on a single axiom: *observation occurs*. Part I examines that phrase directly and finds it resolves into two axioms, not one — an indubitable claim that tokened differentiation occurs, and a substantive further claim that the differentiation recurs. Six of the framework's apparent further assumptions are recovered as lemmas of the first axiom; the second axiom is shown, by a no-go result, not to be derivable from the first. The methodological point is the discipline this exhibits: a foundation stated at the exact granularity of what it commits to, with the single non-evidential posit named and isolated. One element of the distillation is a commitment made explicit where it arises (§5): the observer's embeddedness — the proper-part decomposition $V \subsetneq S$ — is taken to belong to the primitive itself, because the framework reads "tokened differentiation" perspectivally (a registering from a locus, against a remainder). The thin alternative, on which embeddedness must instead be added as a third axiom, is noted there; the choice is presentational rather than empirical, and the two-axiom presentation reflects the adopted perspectival reading.

Part II — constructive structural realism. Ontic structural realism holds that physical structure, not individual objects, is what is fundamental and what is known. As usually stated it is a programmatic position. The framework makes a **scoped** version constructive: the local propagating lattice/gauge residue is shown to be uniquely fixed under the reconstruction theorem's stated conditions, and the residual gauge freedom of the reconstructed dilation/local datum is explicitly exhibited, with the completeness of its classification open. A Bell-inclusive full completion is not part of that uniqueness theorem: H-Bell conditions its existence and its uniqueness is open.

Part III — per-theorem checks of no-go results. The no-go theorems of quantum foundations — Bell, Kochen–Specker, the reality theorems, Frauchiger–Renner — constrain distinct premises. The framework uses its substratum, reduced description and trace-out to locate each premise, and checks the obligations separately. Bell requires ontic parameter dependence in the adopted measurement-independent branch, with H-Bell still open; the other

cases retain their stated operational, preparation and observer-consistency assumptions.

Part IV — physics modulo gauge. The three preceding commitments combine into a general proposal. The realism debate asks what is fundamental; the framework's posture is that this question is ill-posed, and that the well-posed replacement is: what is the equivalence class of structures that produce the observations, and what are its representatives? Part IV articulates this as a methodology for fundamental physics generally, not as a feature peculiar to the framework's particular substratum.

Part V — the operational methodology. Parts II–IV establish physics modulo gauge as the right form of the answer; Part V develops what the methodology determines operationally. Two further no-go results sharpen the boundary between substrate-derivable content and content selected by empirical input: a gauge-invariance no-go on substrate-only computation, and a category-error finding on substrate-level Lagrangians. These two results concern the same substratum and observable descriptions as Part III, with their own gauge-invariance and definability premises. Systematic audits applying the operational standard to the framework's own content are summarized, and three methodological refinements canonized by the audit experience are stated at the generality the methodology requires.

1.3 Developmental status, stated once

The five parts differ in maturity, and it is more honest to say so here than to qualify every paragraph. Part I is a recent distillation; its central no-go claim has been stress-tested but not yet refereed, and it is presented for exactly that scrutiny. Part II summarizes structural results — the uniqueness and gauge-classification theorems — established in the framework's technical papers, at the fidelity of faithful synthesis; a reader who needs the theorem statements themselves should consult those papers. Part III's no-go analysis treats four cases, each to the scope Part III states: two — Bell and Frauchiger–Renner — treated in the framework's technical papers, and two — Kochen–Specker and Pusey–Barrett–Rudolph — developed in this paper (Section 14), each as a chain of reasoning running on results the technical papers already establish. Part IV is a methodological proposal and is argued as such, not asserted. Part V is the most recent material; its two no-go results are developed in the framework's technical record, the systematic audits it describes are developed there as well, and the comparison of these results with Part III's distinct premise checks — together with the three operational refinements at the generality the methodology requires — is the present paper's synthesis, offered as a comparison of distinct arguments with their hypotheses kept separate. None of this paper's content is a derivation of physics; all of it concerns how the framework's physics is organized, and the claims of this paper stand or fall as methodology.

One status distinction inside the organized physics is worth stating here, because

the parts are easy to run together. On the bridge between embedded observation and quantum mechanics, three things hold at three different strengths. Exact finite observable-law representability is **established**: every finite-horizon visible law has a finite reversible realization and a fixed-basis unitary/Born representation, and the three descriptions are equivalent. Exact finite operational quantum mechanics is **characterized conditionally**: within complex matrix kinematics taken as given, it is equivalent to the framework's core together with explicit added principles, none of which bare observation incompleteness supplies, and a derivation of that kinematics is open. Selection of a unique relative quantum evolution from the coherent lifts compatible with one visible family is **open**: ordinary coherence is proved insufficient, no mechanism is endorsed, and nothing in the framework claims the selection is impossible.

PART I — THE FOUNDATIONAL COMMITMENT

Part I develops the framework's foundational commitment: the distillation of "observation occurs" into a two-axiom base, the recovery of six apparent assumptions as lemmas, and the no-go result that isolates the single non-evidential posit. It is the first of the paper's four methodological commitments — the discipline of stating a foundation at the exact granularity of what it commits to.

2. THE EVIDENCE

2.1 What cannot be doubted

The framework invokes the Cartesian move: there is a fact that cannot be doubted, because the act of doubting it instantiates it. It is worth stating that fact in its most compressed form, because the compression is not the form the framework's definitions use, and the gap between them is the subject of this note.

The indubitable fact is **not** "I exist." "I" is a substantial noun; it imports a self, a persisting subject, a bearer of properties. It is not "an observer exists," for the same reason — "observer" is a relational substance-noun, importing both a thing that observes and a persistence through which it does so. It is not "the world exists" — that imports a totality.

What cannot be doubted, stated without importing any of these, is that **there is differentiated content**. There is some *this rather than that*. The registered content, whatever else is true or false of it, is not blank, not homogeneous: it has distinction in it. We will call this *tokened differentiation* — "differentiation" because the content is differentiated, distinction-bearing; "tokened" because what is in question is a concrete occurrence, an instance, not the abstract truth

of a proposition. The abstract fact that seven differs from eight does not *occur*; what occurs is the concrete registering of a this-not-that.

This is the evidence. It is genuinely indubitable: to doubt it is to entertain the doubted against the affirmed, which is to register a distinction, which is an instance of the very thing doubted. It is also genuinely *thin* — thinner than "I exist," thinner than "a world exists." It asserts neither a subject nor a totality. It asserts that differentiation has occurred.

2.2 Why not thinner, why not thicker

One might propose a thinner floor still: "something exists." This is in fact *too* thin to be the evidence. Bare undifferentiated existence is not what is ever presented. What is presented is always some *this as against that* — structured content, not featureless being. Remove the differentiation and nothing remains to be aware of or to doubt. "Something exists" understates the evidence: the evidence includes the differentiation, not merely the existence.

One might propose a thicker floor: "the universe exists," "matter exists," "I exist." Each of these is more than the evidence. Each becomes the object of a coherent doubt the moment one asks how it is known; each, doubted, unravels to the thin fact beneath — that there is differentiated content. The thicker claims are not evidence. They are hypotheses about what lies behind the evidence.

Tokened differentiation is therefore the unique level that is *both* indubitable *and* the actual content of what is given: more than bare existence, less than a world, silent on every question of what underlies it. That it is also exactly thick enough to generate structure — as Section 4 shows — is not built in here; it is a fact established downstream, and it is what makes this level the right floor rather than an arbitrary one.

2.3 What the evidence is silent on

It is as important to state what tokened differentiation does *not* deliver as what it does.

It does not deliver a subject: there is differentiation, but the evidence does not say it is differentiation *for someone*, registered *by* an entity that has it.

It does not deliver a totality: there is differentiation, but the evidence does not say it is part of a larger whole, or that there is anything beyond it.

It does not deliver a ground: there is differentiation, but the evidence is silent on whether it is self-grounded, grounded in something else, or grounded in nothing. It asserts the occurrence and stops.

And — the point that will occupy Sections 3 and 5 — it does not deliver *time*. The fact "there is differentiated content" is, as stated, tenseless. It is satisfied by a single differentiated tableau: a this-here against a that-there, with no before

and no after. The evidence gives a frame; it does not, on its own, give a sequence of frames.

2.4 The distinction-primitive: lineage and the divergence from it

Taking distinction as the primitive is not new, and the note should be explicit about its lineage. The major precedent is George Spencer-Brown's *Laws of Form* [SpencerBrown], which takes distinction as the irreducible starting point — "we cannot make an indication without drawing a distinction," and so "the form of distinction" is taken "for the form" itself. Spencer-Brown treats distinction as a proto-operation prior to all calculation and all representation: one cannot even compute $1 + 1$ without first distinguishing the symbols. The note's Axiom 1 is, in its core, the same recognition — that differentiation is the floor beneath which analysis does not reach.

But there is a deliberate and load-bearing divergence, and stating it precisely is part of what Axiom 1 contributes. Spencer-Brown's distinction is an *act*: the calculus begins with the *drawing* of a distinction, the *making* of a mark; in his treatment, as commentators note, there is no separation between the result of distinction and the *process* of distinguishing. That actival, processual reading is exactly what this note's Section 5.2 isolates as declined-attempt (1) — "distinguishing as act" — and rejects, because an act takes time and so smuggles succession into what is meant to be the tenseless floor. Axiom 1 therefore adopts Spencer-Brown's distinction-as-primitive but in a *tenseless, non-actival* form: it asserts that a distinction is *tokened* — concretely instantiated — not that a distinction is *drawn*. The difference is not cosmetic. It is precisely the difference that keeps Axiom 1 free of the temporal commitment that belongs, instead, to Axiom 2. Where Spencer-Brown's single actival primitive already contains the seed of process, the note's two-axiom structure separates the tenseless content (Axiom 1) from the recurrence (Axiom 2) that an act would have fused into it.

This divergence also marks the limit of the borrowing. *Laws of Form* has been read, in the literature it generated, as a basis for claims about cybernetics, consciousness, and non-dualist metaphysics. The note takes from Spencer-Brown only the formal point — distinction as the irreducible primitive — and, consistent with Section 2.3, remains silent on the metaphysical and phenomenological superstructure that others have built on it. Axiom 1 asserts that a tokened differentiation occurs; it does not assert what the literature around *Laws of Form* has often gone on to assert.

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3. THE EIGHT-RUNG LADDER

3.1 The framework's definitions, unpacked

The framework formalizes "observation occurs" through a definition: an observation is a triple (S, φ, V) — a total system S, a dynamics $\varphi : S \to S$, and an

observer $V \subsetneq S$, a proper subsystem with finitely many distinguishable internal states, coupled to its complement through φ ([Main] §1.2). Three lemmas and four conditions (C1–C4) are then attached. Laid against the evidence of Section 2, this formalization is seen to assert considerably more than "there is differentiated content." The surplus can be itemized as a ladder, each rung a distinct claim:

1. **A distinction occurs.** Differentiated content; this-not-that. *(The evidence of Section 2.)*
2. **The distinction is relational.** It holds between relata — a this *and* a that.
3. **The relata compose a whole the distinction does not exhaust.** There is an S, and the observer-locus V is a *proper* part of it: $V \subsetneq S$.
4. **There is more than one moment.** Temporal succession exists.
5. **A rule connects the moments.** There is a dynamics φ.
6. **The rule is bijective.** φ is deterministic (injective) and reversible (surjective).
7. **The states are finite in number.** The visible configuration space is finite.
8. **The hidden-sector conditions hold.** Non-zero coupling, record persistence (the slow bath one realization), sufficient capacity, history readback (C1–C4).

The framework's theorem — that an embedded observer's finite observable law has an exact fixed-basis unitary/Born representation, Markov laws included, and carries positive conditional memory at an accessible order exactly when history readback holds there, the compressed representation holding for processes in the correspondence's class under (T) — takes all eight rungs as its basis. The evidence supplies only rung 1. The question of Sections 4 and 5 is which of rungs 2–8 are *consequences* of rung 1 and which are *additional commitments*.

3.2 The standard of derivation

A rung counts as derived — as a lemma rather than an addition — only if it follows from rung 1 by an argument that does not, at any step, presuppose a higher rung. The constraint is strict because the rungs are not independent: it would be circular to derive rung 4 using rung 5, or rung 6 using rung 7. Each derivation below is checked against this standard, and where a derivation consumes a premise not yet established, that is recorded as a finding rather than glossed.

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4. WHICH RUNGS ARE LEMMAS

4.1 Rung 2 — relationality

A distinction, by what the word denotes, holds between relata. A distinction that distinguishes nothing from nothing is not a distinction; the notion is incoherent without at least two terms. Rung 2 is therefore not an addition to rung 1 but an unpacking of it: "differentiation" already means *differentiation of something from something*. Rung 2 collapses into rung 1 by conceptual analysis.

4.2 The locus, and rung 3 — the proper part

Rung 3 asserts that the observer-locus is a proper part of an unexhausted whole. Between rung 2 and rung 3 there is a step easy to miss: rung 2 gives two relata symmetrically, while rung 3 gives an asymmetric part-whole structure, one relatum *contained in* a whole it does not exhaust. The step requires a premise — call it the *locus* premise — that the differentiation is registered *at* a locus, a where of the registering distinct from the what registered.

The locus premise is not a free addition. It is recovered from the verb in rung 1. To say a distinction *occurs* — as against merely *obtaining*, the way an abstract mathematical inequality obtains — is to say it is instantiated as a concrete token. And a concrete token is located: occurrence is indexical, a token is a *this-here*, instantiation places. The locus premise is therefore analytic in "occurs," just as relationality is analytic in "distinction." Once the locus is granted, rung 3 follows: the locus V, together with the registered content beyond it, composes a whole S of which V is a proper part, because V does not include the content beyond it. Rung 3 is a lemma.

This recovery has a substantive consequence. The proper-part structure $V \subsetneq S$ — the *incompleteness* the framework is named for — is not an independent postulate. It is a consequence of the evidence, provided "occurs" is read as "is tokened." Embeddedness is analytic in occurrence.

This recovery rests on a reading of the primitive, adopted here explicitly. The step from the locus premise to rung 3 depends on reading "tokened" as carrying a *perspectival* locus — a registering *from* a within-place, against a remainder. On a thinner reading, "tokened" means only that the differentiation is concretely instantiated, with no within-place; on that reading the locus premise does not yield a *proper* part, and $V \subsetneq S$ would be a third axiom rather than analytic in the primitive. The framework adopts the perspectival reading — the floor the cogito secures is already a registering — so embeddedness is part of the primitive and the foundation is two axioms, not three. The choice is presentational: it changes no result downstream of the structure (S, φ, V). It is stated here rather than deferred, and rung 3 is presented as analytic in the perspectival primitive, with the thin-reading alternative noted.

4.3 Rung 5 — the existence of a dynamics

Granting (provisionally, pending Section 5) that there is more than one moment, and granting the locus that persists across moments as the same locus, there is necessarily *something* that relates the configuration at one moment to the configuration at the next. That relation is the dynamics φ. Rung 5 — the *existence* of a dynamics — is therefore a lemma, given rung 4 and the locus.

Two cautions. First, the lemma delivers only the existence of a successor relation, not its uniformity: that the same φ governs every locus and every moment is a further claim, recovered downstream from isotropy and homogeneity, not here. Second, the lemma depends on rung 4, which Section 5 will show is *not* itself a lemma. Rung 5 is thus a *conditional* lemma: a consequence of the evidence together with the one genuine addition.

4.4 Rung 6 — bijectivity

The framework's Lemma 3 derives bijectivity from the preservation of the observer's records. The derivation separates into two halves with different costs.

Determinism (injectivity). A registered distinction, once it has occurred, has occurred. If the dynamics φ merged two distinct total states into one successor — if φ were non-injective — then two configurations carrying distinct registered records would become a single configuration, and a distinction that had occurred would cease to have occurred. Rung 1 asserts the distinction occurs; persistence of the record across moments (rung 4 plus the locus) carries that occurrence forward; non-injective φ would violate it. So φ is injective. Determinism is a lemma, given rung 1, rung 4, and the locus.

The hidden-sector margin, closed. The record argument as just stated covers merges of configurations carrying distinct registered records. A merge of configurations differing *only in the hidden sector* erases no record, and needs a separate argument; there is one, and it is a dichotomy. If the merged configurations' successors differ visibly, the merge leaves a statistical trace: the marginal of a bijection under the uniform prior is doubly stochastic (by counting: each visible value has exactly as many preimages as there are hidden states; the framework's Main paper, §3.2), unitary emergent statistics are unistochastic, and a visibly-asymmetric merge breaks double stochasticity of the marginalized process at some step order — so such merges are excluded where the observed statistics are of that doubly stochastic class under the uniform prior, an evidential input rather than a further axiom. If instead the merge leaves no statistical trace, it is removable: any map on a finite set restricts to a bijection on its eventual image, a recurrent registered history lives in that image, and the non-injective description is gauge-equivalent to a bijective one — the merge is a pre-history transient with no observational content, and the bijective representative is canonical (this is Part IV's surplus-as-inaccessibility, applied to the dynamics itself). Either horn delivers the bijective substratum for the dynamics governing registered history. The honest accounting, which the derivation's pu-

rity label should carry: injectivity on record-distinct configurations is a lemma of rung 1; injectivity on the hidden-sector margin is delivered by observed unitarity or by recurrence — a disciplined mixture of the evidential floor with one empirical input, not analysis of the evidential floor alone.

Reversibility (surjectivity). Injectivity yields bijectivity only on a finite set; on an infinite set an injective map need not be surjective. Reversibility therefore is not established here — it waits on rung 7.

4.5 Rung 7 — finiteness

Rung 7 asserts the visible configuration space is finite. The framework's Lemma 1 motivates finiteness from holographic entropy bounds — but those are downstream physical results, and invoking them at the foundational layer would be circular. The question is whether finiteness follows from rung 1 alone.

It does, in the form the framework actually requires. A *token* is a determinate concrete particular. A single tokened differentiation registers a determinate, and therefore finitely structured, content: an *infinite completed totality* of tokened distinctions is not something that can *occur*, because occurrence is the instantiation of a determinate particular and a completed infinity is not a determinate particular. Infinitely many distinctions could be *constructed* — one tokening after another, across moments — but never *given* in a single occurrence. The evidence therefore delivers finiteness *per moment*: each occurrence registers finitely many distinctions.

Finiteness-per-moment is exactly what the observation base delivers, and the scope of what it delivers should be stated on both of its axes. On the time axis: the visible configuration space $\mathcal{C}_V$ whose finiteness Lemma 1 requires is the space of states distinguishable *at the locus*, a per-moment object; the framework does not require, and should not claim, finiteness of the entire history, and an unbounded number of moments is consistent with the evidence. On the space axis: the tokening argument reaches only the *visible* factor — what the observer registers — so what is recovered as a lemma is per-moment finiteness of $\mathcal{C}_V$. The substratum arguments that act on the *total* per-moment space $\mathcal{C}_V \times \mathcal{C}_H$ — the exact-recurrence step and the eventual-image restriction of the determinism recovery — consume finiteness of the hidden factor as well, and that extension is not delivered here: it is the structural assumption A1 of the substratum construction, counted as such in the companion paper's posit ledger ([Main §4.5], of which this rung analysis is the complementary view — the ladder counts what is recovered from the two-axiom base, the ledger counts what the full argument assumes). Rung 7, in the form Lemma 1 needs, is a lemma.

This completes rung 6 with the same scoping. Determinism (4.4) plus finiteness-per-moment yields bijectivity as a lemma on the visible factor: an injective map on a finite per-moment configuration space is surjective on it. Bijectivity of the substratum map on the total space inherits the A1 extension above; reversibility of the registered dynamics is recovered, and rung 6 is a lemma at the visible

level, structural-plus-A1 at the substratum level.

4.6 Rung 8 — the hidden-sector conditions

Rungs C1–C4 are conditions on the relationship between the locus V and its complement. C1, non-zero coupling, follows conditionally: a distinction with content *about* the complement requires coupling to the complement, on pain of that content being unregisterable. C2 (record persistence — slow-bath separation one route) and C3 (sufficient capacity) are not lemmas — they are substantive conditions satisfied by some partitions and not others. The framework does not claim them as lemmas: it presents C1–C4 as the conditions defining *which* embeddings carry the memory-bearing sector — hidden predictive memory unavoidable in every completion, the quantum representation itself universal ([Main] §3.4) and as a structural selection rule on where observers can exist ([Main] §4.6). Rung 8 is correctly treated by the framework as a selection condition, not a derivation, and no overclaim is present.

4.7 Summary of the recoveries

Of rungs 2–8: rung 2 (relationality) and the locus premise collapse into rung 1 by conceptual analysis; rung 3 (proper part), rung 5 (dynamics, conditionally), rung 6 (bijectivity), and rung 7 (finiteness-per-moment) are lemmas; rung 8 (C1–C4) is a selection condition the framework already treats as such. Exactly one rung remains: rung 4.

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5. THE ONE RUNG THAT IS NOT A LEMMA

5.1 Two questions about succession, and which one is the claim

The claim of this section concerns rung 4 — temporal succession. But "deriving succession" can mean two distinct things, and the no-go applies to one and not the other. The distinction is essential and the rest of this note depends on it.

The first question is the **ordering** question: *given* that there are multiple moments — multiple states of some persisting thing — can the relation "before/after" among them be defined without circularly invoking "before"? This is the problem the philosophy of time has worked on for a century. It is contested but not obviously hopeless: a reductive account of the ordering relation may well be possible (Section 5.3 discusses a recent candidate). The no-go of this note is *not* a claim about the ordering question.

The second question is the **domain-existence** question: is there a temporal domain *at all* — is there more than one moment, more than one state of a persisting thing — and can *that* be derived from a premise which does not already contain it? This is the question the no-go addresses, and the answer is negative.

The distinction matters because the two questions are routinely conflated, and an account that answers the first is easily mistaken for one that answers the second. An account of the ordering relation always *begins* from a set of moments, or a set of states of an object, or a configuration space — a multiplicity given at the outset — and derives the order *on* that multiplicity. It does not derive the multiplicity. The multiplicity is its input. And the multiplicity — that differentiation is instantiated across moments and not only within a single configuration — is exactly what the framework's second axiom asserts (Section 5.4).

Rung 4, as the framework needs it, is the domain-existence claim: that there is a temporal domain for the dynamics φ to act across. The claim of this section is that this cannot be derived from rung 1. Tokened differentiation, as established in Section 2, is tenseless: it is satisfied by a single differentiated tableau — a configuration, one frame — with no before and no after. To reach even *two* moments from a single differentiated frame requires a premise the frame does not contain. The argument takes the form of a small no-go result.

The two-question distinction also fixes the complete inventory of the framework's temporal commitments. The framework's downstream physics involves more temporal structure than the bare domain-existence claim — it involves an *ordering* of moments, a *causal topology*, and an *arrow* of time — and a reader is owed an account of how these relate to the two axioms. The inventory has four levels. *Multiplicity* — that there is more than one moment — is the domain-existence claim; it is the framework's second axiom. *Ordering* — that the moments form a sequence — is the ordering question above; the framework leaves it open. *Causal topology* — that the ordering carries influence-structure, that earlier moments can affect later ones — is what the dynamics requires to be stated at all. *The arrow* — that the ordering has an experienced direction — is addressed in the framework's substratum treatment: on an expanding homogeneous general-relativistic background satisfying the null energy condition, the apparent-horizon area is a non-decreasing parameter (background kinematics), and the direction rests on boundary conditions and the orientation convention. The relations among these four are definite, and reduce the inventory to the two axioms-or-questions already named. The causal topology is not an independent commitment: an influence-structure on an ordered set of moments connected by φ is nothing over and above the ordering together with φ itself, since "earlier influences later" just means "related by forward iteration of φ"; the dynamics φ is independently present, so the causal topology reduces to the ordering. The arrow rests on the causal topology, hence on the ordering, and on those boundary conditions. So the framework's genuinely independent temporal commitments are exactly two — multiplicity and ordering — and they are precisely the two questions this section distinguishes: multiplicity is settled by the second axiom, ordering is left open. Causal topology and the arrow are downstream of these two and of inputs the framework takes as given (the dynamics, the expanding branch, the boundary conditions that fix the arrow's direction). No third temporal axiom is required: the two-question framing of this section accounts for the

framework's temporal axioms, and the arrow's boundary conditions are inputs rather than axioms.

5.2 Six attempted derivations

The claim is established not by a single argument but by the exhaustion of the available derivations. Six attempts to derive succession from differentiation were constructed; each fails, and — the point of substance — each fails at a *distinct, identifiable step*, always the same kind of step: the illicit move from a state-predicate to a process-predicate, or from logical order to temporal order.

1. *Distinguishing as act.* "To distinguish is to do something, and doing takes time." Fails: it reads "distinguish" as an act, a process, when the evidence gives only differentiated content statically present. The frame need not be *arrived at*.

2. *Registering as record.* "To register is to inscribe a record, which persists to be read; reading is later than inscribing." Fails: it builds persistence — succession itself — into the meaning of "register." If "register" means only "differentiated content is present," no second moment follows.

3. *Actualizing as transition.* "To be actual rather than possible is to have been actualized, a transition from possibility to actuality." Fails: "transition" and "actualized" are temporal. Actuality can simply *be the case* tenselessly; the space of unrealized alternatives is a contrastive description, not a prior state the token passed through.

4. *Occurrence as actualization.* A refinement of (3) taking "occurrence" as the primitive. Fails at the same step: it still requires actuality to be the *outcome* of an actualizing, and "outcome" is temporal.

5. *Occurrence taking itself as object.* "Occurrence occurs — the occurrence of occurrence — and this self-application unrolls into a sequence." Fails differently: self-application generates a *logical* hierarchy of orders, not a *temporal* sequence. Reading the logical "of" as a temporal "then" is the added premise. (This attempt has a second defect: it imports a self-grounding, *causa sui*, thesis not present in the evidence, trading evidential austerity for elegance.)

6. *A single fused term.* The proposal that some sufficiently broad word denotes axiom and succession together as one. Fails: a single term denoting two separable contents does not unify them; it conceals the seam, so that subsequent users inherit the succession commitment without being told they have made it. Concealment is not derivation.

Six attempts, six distinct points of failure, one invariant: succession is reintroduced, never derived. The consistency of the failure across genuinely different framings is the evidence that the obstruction is structural — a fact about what can be inferred from what — and not an artifact of any one bad formulation.

A static premise does not yield a process conclusion; "differentiated content is" does not yield "differentiated content comes to be." In each case what cannot be reached is the domain-existence claim of Section 5.1: each attempt, to get a second moment, must assume a second moment.

5.3 The result against the existing literature

The no-go of Section 5.2 is not made in isolation from a substantial existing literature on temporal succession. That literature both corroborates the result and sharpens the distinction of Section 5.1.

Corroboration — Whitrow. G. J. Whitrow, in *The Natural Philosophy of Time* [Whitrow], surveyed the causal theorists' attempts to define temporal succession — the ordering of events by "before" and "after" — and found that they failed because they "tacitly invoked the very temporal order to be defined." Whitrow's verdict on Reichenbach's mark method, among the most ingenious such attempts, was the same. He concluded that "before" and "after" must be accepted as elementary. "Tacitly invoked" is precisely the diagnosis of Section 5.2: the attempts smuggle. That a major figure in the philosophy of time reached the same finding by independent survey is evidence the obstruction is real and not an artifact of this note's particular six framings.

The ordering question may be tractable — and that does not touch the no-go. It would overstate the result to claim the *ordering* relation cannot be reductively defined. A recent account (Wassermann [Wassermann]) proposes deriving "before" from a timeless "existence asymmetry" on the set of an object's realized states — a genuine candidate for answering the ordering question non-circularly. But the construction begins from M_X, the set of an object's actually-realized states. A set of multiple states of one persisting object is already a temporal domain: one object, many states, is exactly domain multiplicity. The account derives the *ordering* of those states; it assumes the *multiplicity*. This is not a criticism of the account — it answers the ordering question, which is the question it set out to answer. It is an illustration of the Section 5.1 distinction: even the most sophisticated reductive account of "before" takes the multiplicity as its starting set and cannot, by its own construction, derive it.

The same pattern in physics. The timeless approaches to quantum gravity exhibit the identical structure. Barbour's timeless physics [Barbour] takes "possible configurations of the universe and a law" as fundamental and derives time — but the configuration space, the multiplicity of possible instants, is primitive; what is derived is the ordering and the arrow. The Wheeler-DeWitt equation is timeless and time is recovered semiclassically — but on a space of geometries given at the outset. Even recent attempts to derive time from quantum entanglement have been observed to "use time in the set-up." Across both literatures — philosophy of time and the physics of timeless approaches — the same result holds: accounts derive the ordering of moments, or the arrow among them, from structures that already contain the multiplicity of moments. None derives the

multiplicity. The domain-existence question is, in the words this note's Section 5.2 reached independently, not one that any known account answers.

Difference and repetition — Deleuze. The pairing of this note's two axioms — differentiation, and its recurrence — is the explicit subject of Deleuze's *Difference and Repetition* [Deleuze], the major modern treatment of the difference/repetition pair. The engagement matters because Deleuze's account of repetition is, at first sight, in tension with this note's Axiom 2. Deleuze insists repetition is "difference without a concept" — *not* the recurrence of the same but the production of novelty; and he ties repetition to the synthesis of time, citing Hume's observation that a repeated series (AB, AB, AB…) itself produces something new, a habit, an expectation. One might read this as a claim that recurrence is not a primitive but a synthesis, and that time is its product — which would make Axiom 2 a derived structure rather than an axiom. It does not. Deleuze's syntheses operate on an *already-given manifold* of instants: the AB,AB series already has its multiple terms; the passive synthesis of habit contracts a succession that is already there. What Deleuze derives — habit, expectation, the directedness from past through present to future — is temporal *ordering and arrow*, the Section 5.1 ordering level. He does not derive, and does not claim to derive, the bare existence of more than one instant. Deleuzian repetition therefore stands with Wassermann and Barbour: it presupposes the multiplicity and works upon it. It corroborates the no-go rather than threatening it. Two points of contrast are worth stating plainly. First, this note uses "recurs" in Axiom 2 in a deliberately thinner sense than Deleuze's "repetition": Axiom 2 asserts only that differentiation is instantiated more than once over, taking no position on whether the recurrence reproduces the same or produces novelty — that richer question is downstream of the bare domain-existence Axiom 2 asserts. Second, where Deleuze makes difference the sole ontological primitive and repetition a synthesis, this note finds two posits, not one, precisely because — as Section 5.2 establishes — the recurrence cannot be synthesized from the difference without a premise that already contains it.

5.4 The status of succession — a second axiom

Succession is therefore neither evidential nor derivable. But it is also not an independent axiom of the same kind as rung 1. The honest characterization is more specific.

Differentiation can be instantiated more than once over. There is differentiation *within* a configuration — this-here against that-there, the content of a single frame. And there is differentiation *across moments* — this-moment against that-moment. The principle invoked is the same in both: tokened differentiation. What the second instance adds is not a new principle but a new fact — that the principle is instantiated *recurrently*, more than once over, and not only within a single frame.

That fact is a second axiom. It is tempting to call it a mere parameter — a

setting on the one axiom — or to absorb it by saying the one axiom is simply "applied across multiple domains." Both moves fail, and fail informatively. A posit that can be neither derived nor eliminated, and must be asserted, is an axiom, whatever it is called; "parameter" understates it. And "applied across multiple domains" is grammatically singular but contentually dual: application presupposes the existence of what one applies across, so the phrase silently carries the very second posit it appears to dissolve. The honest count is two.

The two axioms are:

- **Axiom 1.** *Tokened differentiation occurs.* Concrete differentiated content is instantiated; there is some this-not-that. This is the evidence of Section 2. It is indubitable — its denial is itself a tokened differentiation — and it is conceptually primitive: even the vocabulary needed to state the second axiom presupposes it.
- **Axiom 2.** *Differentiation recurs.* The principle of Axiom 1 is instantiated more than once over — not only within a single configuration but across a further axis. The framework's temporal domain is the minimal such recurrence; the configurational and temporal instances are the two minimum witnesses, and the actual count of axes is settled downstream by derivation, not fixed here. What Axiom 2 asserts is bare recurrence — that the principle holds more than once over — and nothing more: it does not by itself supply an ordering of the recurrent instances, a causal topology, or the specific dynamics φ; these are either left open (ordering, §5.1), derived downstream from structure the framework independently has (causal topology, §5.2 inventory), or addressed downstream with boundary conditions taken as inputs (arrow, §5.2 inventory). The floor is two posits at the granularity stated, and the dynamical content the downstream physics requires enters through later, separately-stated structural conditions rather than through Axiom 2's semantics.

The two axioms stand in a definite relationship, and stating it precisely is part of the result. Axiom 2 **presupposes** Axiom 1: there can be no recurrence of differentiation without differentiation, and the very notion of an "axis" or "domain" along which the recurrence runs is itself a differentiation-structure, unstatable without Axiom 1's content. But Axiom 2 is **not derivable from** Axiom 1: that a differentiation occurs does not yield that it recurs — this is the no-go of Section 5.5. The relationship is therefore one-directional presupposition without entailment. Axiom 1 is first — conceptually and epistemically. Axiom 2 is genuinely additional. They are two, ordered, and not reducible to one.

A note on what is *not* claimed. The two axioms are not unrelated posits. The principle they invoke — tokened differentiation — is single; Axiom 2 is that single principle asserted to recur. The unity is real, but it is a unity of *theme*, not of *count*: one principle, two axioms. To let the thematic unity be reported as a unity of count — "one axiom" — would be to conceal the second posit behind the first, which is the move this distillation exists to expose, not to perform.

5.5 The result stated as a no-go

The content of Sections 4 and 5 can be stated as a single result. *Given Axiom 1, "tokened differentiation occurs," the rungs relationality, proper-part, dynamics-existence, determinism, finiteness-per-moment, and bijectivity follow as lemmas; the recurrence of differentiation — Axiom 2, equivalently the existence of more than one moment for the dynamics to act across — does not, and cannot, follow.* The negative half is a no-go theorem of modest scope but real content: it identifies, exactly, the one place where the framework's foundation makes a second commitment, and it shows that the second commitment cannot be discharged by any derivation from the first — it must be asserted as an independent axiom. The no-go is specifically about the *recurrence* of differentiation (the existence of the temporal domain), not about the *ordering* of moments within that domain; the latter is a separate and possibly tractable question (Section 5.3), and the framework takes no position on it here.

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6. THE TWO-AXIOM OBSERVATION BASE

6.1 Statement

The foundation of the framework, distilled, is two axioms:

> **Axiom 1.** *Tokened differentiation occurs.* — Concrete differentiated content is instantiated: there is some this-not-that. Indubitable; conceptually primitive.
>
> **Axiom 2.** *Differentiation recurs.* — The principle of Axiom 1 is instantiated more than once over, across a further axis. The temporal domain is the minimal such recurrence; configurational and temporal instances are the two minimum witnesses, the actual count of axes being settled downstream by derivation.

We call this two-part foundation the **two-axiom observation base**. The two axioms are ordered: Axiom 2 presupposes Axiom 1 (recurrence of differentiation requires differentiation, and the very notion of an axis along which recurrence runs is a differentiation-structure), but Axiom 2 is not derivable from Axiom 1 (Section 5.5, the no-go). The relationship is one-directional presupposition without entailment.

The foundation comprises two axioms, not one axiom plus a domain-multiplicity *parameter*: a posit that can be neither derived nor eliminated, and must be asserted, is an axiom, so the count is two. The two axioms invoke a *single principle* (tokened differentiation), Axiom 2 being that principle asserted to recur. The unity is thematic, not numerical — one principle, two axioms — and is recorded as the relationship between the axioms, not as a reduction of their count.

Everything else in the foundational layer is a lemma: relationality, the locus, the proper-part structure $V \subsetneq S$, the existence of a dynamics, determinism, finiteness-per-moment, and bijectivity. The hidden-sector conditions C1–C4 are selection conditions, not lemmas, and are so treated.

6.2 What the two-axiom form buys

The two-axiom statement is more honest than "a single axiom" and more precise than "several assumptions," and both improvements are substantive.

Against "a single axiom": that phrasing, unqualified, suggests the framework's foundation is a single indubitable fact. It is not. It is one indubitable fact (Axiom 1) *plus* one further commitment (Axiom 2) that is not indubitable — a world without recurrence, a single tenseless configuration, is coherently conceivable, merely a world with no dynamics and no physics. The two-axiom form states this. It does not weaken the framework: a foundation whose second, non-indubitable commitment is named, isolated, and shown by a no-go theorem to be irreducible to the first is more defensible than a foundation whose commitments are distributed invisibly through a definition. The framework need not claim "one axiom"; the true claim — two axioms, ordered by presupposition, provably not collapsible — is the stronger thing to be able to say, because it comes with a theorem rather than a slogan.

Against "several assumptions": the eight-rung ladder might suggest the framework rests on eight independent posits. It does not. Six of the eight are lemmas. The framework rests on two axioms, and the two-axiom form states *that* — the economy is real, and naming it at the correct count is part of stating it honestly.

6.3 Relation to other minimal claims

The two-axiom observation base is sometimes compared, in discussion, to other candidate "bedrock" claims. The comparisons are clarifying. They concern Axiom 1 — the indubitable, conceptually primitive member — since it is Axiom 1 that occupies the "bedrock" role.

It is not "the universe exists." That asserts a totality; Axiom 1 asserts only tokened differentiation, and the totality (S, the unexhausted whole) is recovered as a lemma (rung 3), not assumed.

It is not "a being exists" — in particular not any claim of the form "X exists" for X a substance with properties. The distillation refuses substance-nouns throughout: it rejected "I exist" and "an observer exists" for importing a persisting subject, and rejected "occurrence grounds itself" for importing a self-grounding thesis. Axiom 1 asserts an occurrence and is silent on what, if anything, underlies it. It is ground-neutral by construction, and that neutrality is deliberate: it is the same discipline that rejected attempt (5) of Section 5.2.

It is not "something exists" simpliciter. That is too thin to be the evidence (Section 2.2): bare undifferentiated existence is not what is given. Axiom 1

asserts *differentiated* content — more than bare existence — which is both what is actually given and, as Section 4 shows, exactly enough structure to generate the lemmas.

Axiom 1 belongs to the family of claims that cannot be doubted because doubting instantiates them — the family of the Cartesian *cogito*. It differs from the others in that family by being maximally disciplined about claiming *only* the indubitable floor and not the structure that the tradition has usually built onto it without notice. Axiom 2, by contrast, makes no claim to indubitability; it is a substantive posit, defended not as undeniable but as the minimal further commitment without which there is no dynamics and no physics, and shown by the no-go to be irreducible to Axiom 1.

6.4 Relation to the landscape: generality, selection, and what they presuppose

A natural question, given that the framework elsewhere derives a gauge structure and a selection principle, is whether the two axioms might themselves be understood on the model of the string-theory landscape: the laws are not unique but range over a vast space of consistent possibilities — gauge or moduli freedom in the loose sense — and *which* possibility obtains is fixed, in our case, by anthropic selection, since only observer-permitting possibilities are observed. Could Axiom 2 — the recurrence of differentiation, the existence of the temporal domain — be such a selected feature: gauge in general, specific in our anthropic case?

It could not, and the reason is the same no-go established in Section 5, arriving from a new direction. Anthropic selection on a landscape operates on features *within an already-constituted physics*: the cosmological constant, the compactification, the couplings each take different values across the landscape, but every point of the landscape is a temporally-extended world with a configuration space and a dynamics. The landscape is a landscape *of* differentiated, temporally-extended things. Anthropic selection picks among such worlds; it cannot pick whether there is a temporal domain at all, because the selection apparatus — a measure over possibilities, the counting of observer-permitting cases — already presupposes the temporal domain. A "world" with no recurrence of differentiation has no dynamics, no vacua, nothing to be a point of a landscape and nothing for a selection measure to weigh. The proposal "Axiom 2 is an anthropically-selected landscape variable" therefore relocates the second commitment into the meta-level structure of the landscape rather than removing it: the landscape itself presupposes Axiom 2. This is the same outcome as the "parameter" and "condition" framings of Section 5.4 — a setting is a setting *within* a space of possibilities, and the space of possibilities cannot be constituted without the very structure the setting was meant to make optional. It may be recorded as a further attempt — alongside the six of Section 5.2 — examined and found to presuppose what it would explain.

String theory itself bears on the two axioms only in this presuppositional way. It is formulated on a worldsheet with a time coordinate, in a target space with a time dimension, over a configuration space that is a multiplicity; it inherits both Axiom 1 and Axiom 2 and questions neither. String theory is not a competing account of the foundation discussed here — it operates well downstream of it, and whatever its landscape is, it is a landscape of worlds in which both axioms already hold.

The genuine and useful point in the landscape comparison lies *below* the two axioms, not at them. Downstream of Axiom 1 and Axiom 2, the framework does have a "general, then selected" structure, and it is one of its established features rather than a speculation: the emergent description is gauge — partition-relative, with the substratum gauge group identifying which structure is observable — and the partition that supports an observer is then *selected*, by the structural observer-selection theorem, conditional on the effective-mixing hypothesis (EM) and its rate condition ([Main §4.6]), as one of those satisfying the embedding conditions. This is the framework's analogue of "general law, particular vacuum"; it is the natural counterpart of the landscape-plus-anthropics picture. The difference worth stating is that the framework's version is derived under that hypothesis: where the landscape picture posits a space of vacua it cannot derive and an anthropic measure it cannot rigorously define, the framework's gauge structure is constructively exhibited, with its classification's completeness open, and its selection principle is a theorem, under (EM), about which partitions admit embedded observers. The "gauge in general, specific in our case" intuition is therefore correct and is realized in the framework — but it is realized one level down, among the partitions and the emergent descriptions, not at the two axioms, which are precisely what must hold for there to be a space of partitions to be general over in the first place.

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7. THE INTUITION IN PLAIN LANGUAGE

The preceding sections are precise at the cost of being technical. The underlying intuition is simple, and stating it plainly is worthwhile — both because the simplicity is the point and because a reader who holds the intuition can check the technical claims against it.

Begin with the one thing that cannot be doubted. Not "the world is real," not "I exist," not that anything in particular lies behind appearances — those are already guesses about what is behind it. Strip them away. What remains, before any guessing, is this: there is something rather than nothing, and not a blank something but a something *with a difference in it*. This, not that. The fact that things are not all the same.

That is the whole starting point. Not "a universe exists" — that already assumes one great container holding everything. Not "a God exists" — that assumes a thing that has properties and does things. Just: *difference shows up*.

It is the right floor, and not a thinner or thicker one, for a specific reason. Thinner — "something exists" — is too thin: plain existence with no difference in it is not anything you are ever actually given, and nothing could be aware of it or doubt it. Thicker — "a universe," "matter," "I" — is smuggling: each sounds obvious only because one already lives inside the assumption, and each, once questioned, dissolves back to the thin fact underneath, that there is difference.

Now the second piece, the piece that took the most care to get right. There are two ways difference can show up, and they are genuinely different. Difference *across a scene*: this thing here is not that thing there — two things side by side, in one frozen snapshot. And difference *across time*: this moment is not that moment — one thing, but later.

One might hope the second follows free from the first — that "there is difference" hands you time automatically. It does not. A single frozen photograph is full of difference — light here, dark there, edges, shapes — and contains no time whatsoever; nothing in it is before or after anything else. So "there is difference" gives the photograph. It does not give the movie. To get the movie — time, change, one-thing-then-another — one must add something: that difference runs in the time direction too, not only across the scene. And that addition cannot be squeezed out of the first claim. Six ways of trying to squeeze it were attempted; every one smuggled time in through a word that already had time hidden inside it.

So the honest foundation is two things of two different kinds. The first is *known*: there is difference — it cannot be doubted, because doubting is itself a telling-apart. The second is *assumed*: the difference runs in the time direction as well. The framework does assume the second — plainly our world has time — but it is honest that this is a thing put in, not a thing found.

In one sentence: *"there is difference" is what we know; "the difference includes time" is the one thing we assume; physics is what follows from the two — and being able to point at exactly that one assumed piece is what makes the foundation honest.*

That last clause is the reason the exercise matters. Most attempts at a foundation for physics begin, without noticing, from "there is a world, with things in it, and time passing," and never identify how many separate commitments that involves. The two-axiom observation base says: one thing is indubitable, one further thing is assumed, and here is precisely which is which. A foundation in which the single assumed brick can be pointed at is sturdier than one in which the assumptions are smeared invisibly through everything — not because it assumes less in total, but because it is honest about what it assumes.

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8. OPEN QUESTIONS AND STATUS

Part I is developmental, and several questions are deliberately left open.

The succession no-go. The central negative result — that the existence of the temporal domain is not derivable from tokened differentiation — rests on the exhaustion of six derivation attempts (Section 5.2), corroborated by the independent verdict of the philosophy-of-time literature (Section 5.3). Six is not infinity. The result would be overturned by a seventh attempt that succeeds: a derivation of the temporal domain — more than one moment — from a tenseless differentiation premise that does not, at any step, smuggle a process- or temporal-order predicate, and does not begin from a multiplicity (a set of states, a configuration space) that already contains the domain. The author has not found such a derivation and has given a structural reason to expect none exists — the state-to-process gap is not the kind of gap an argument crosses — but the question is left open for scrutiny. A philosopher of physics is the natural reader to either close it or confirm its standing. The distinction of Section 5.1 should be kept in view in any such scrutiny: a reductive account of the *ordering* of moments (Section 5.3 notes a recent candidate) does not bear on the no-go, which is about domain-existence.

The count, and the relationship between the axioms. Part I settles on two axioms — Axiom 1 (tokened differentiation occurs) and Axiom 2 (differentiation recurs) — related by one-directional presupposition: Axiom 2 presupposes Axiom 1 but is not derivable from it. Earlier formulations that sought a single-axiom statement (a "parameter," a "condition," a principle "applied across domains," a reverse derivation of Axiom 1 from a domain-multiplicity axiom, or an anthropically-selected landscape variable — Section 6.4) were each examined and rejected; each, on inspection, either understated Axiom 2's status or relocated the second commitment without removing it. The two-axiom reading is the current resting point. It is offered, like the rest of Part I, as developmental: whether the relationship between the axioms admits a still more economical regimentation is left open, but no single-axiom formulation found so far survives scrutiny, and the burden is on any proposed one to show it does not conceal the second posit.

What is not open. Two points are not left open. First, the lemma recoveries of Section 4 — relationality, the locus, the proper-part structure, dynamics-existence, determinism, finiteness-per-moment, bijectivity — are presented as established, given the axiom; they turn on conceptual analysis of "occurs" and "tokened" and are checkable directly. Second, the distillation has no bearing on whether the framework's downstream derivations succeed; it concerns only the formulation of their starting point.

PART II — CONSTRUCTIVE STRUCTURAL REALISM

Part II develops the framework's second methodological commitment: that ontic structural realism, ordinarily a programmatic stance, becomes in this framework a constructively demonstrated result. Two theorems carry the claim — a reconstruction theorem and a gauge-classification theorem — and the methodological point is what their conjunction makes possible: a realism about structure that is earned by construction rather than asserted as a philosophical position. The theorem statements summarized here are established in the framework's substratum paper; this part states them at the fidelity of faithful synthesis and refers the reader needing the proofs to that paper.

9. STRUCTURAL REALISM AS A PROGRAMMATIC POSITION

Ontic structural realism (OSR) holds that what is fundamental, and what physical theory gives us knowledge of, is structure — the network of relations — rather than individuals bearing intrinsic natures. The distinction between OSR and a merely epistemic structural realism was drawn by Ladyman [Ladyman1998], and the ontic position has been developed principally by Ladyman and Ross [LadymanRoss] and by French [French2014], with the relation to quantum physics worked out by French and Ladyman [FrenchLadyman2003] and a recent "effective" refinement given by Ladyman and Lorenzetti [Ladyman-Lorenzetti]. The position emerged from two pressures: the pessimistic meta-induction, which observes that the theoretical objects of successive physical theories are repeatedly discarded while their structural-mathematical content is retained; and the underdetermination of object-level descriptions by structure, sharpest in physics where distinct object assignments (particles-as-individuals versus particles-as-modes, for instance) leave all relations and all predictions intact.

OSR responds: take the structure as the ontology, and the recurring instability of the object level becomes unsurprising, because the object level was never the bearer of the theory's content. A standing concern for the position, recognized in its own literature, is that "structure" not collapse into "pure mathematics" — French and Ladyman were explicit that OSR is not committed to the world being purely mathematical — and the question of how the structural and the physical are related remains under discussion. The framework's treatment, developed below, bears on exactly this concern: it makes the structure a specified mathematical object while keeping the physical content fixed by empirical inputs, and it is explicit (Section 12) about what is thereby claimed and what is not.

As ordinarily developed — in the work of Ladyman, French, and others — OSR is a *programmatic* position. It is a stance one adopts about physical theory in

general, defended by general arguments (the meta-induction, the underdetermination), and it makes a general promise: that for a mature physical theory, the structure could in principle be specified and the object-level residue shown to be surplus. The promise is rarely discharged for any specific theory. What "the structure" is, precisely; which respects of a theory's description are structural and which are surplus; whether the structure is uniquely fixed by the empirical content or itself underdetermined — these are, in the programmatic version, left as the position's horizon rather than its deliverable.

The methodological claim of this part is that the Incompleteness framework discharges a **scoped** version of that promise. The local propagating lattice/gauge residue is exhibited and shown unique under Theorem 23's stated conditions, while the surplus of the reconstructed dilation/local datum is exhibited and classified, with exhaustiveness open. Bell-composite structure lies outside that uniqueness result: H-Bell is an existence/compatibility condition and Bell-inclusive uniqueness is open. OSR is therefore constructive here at the local reconstructed level, not a theorem selecting one complete ontology.

10. THE RECONSTRUCTION THEOREM: UNIQUENESS OF THE LOCAL STRUCTURAL RESIDUE

The framework's forward direction runs from a finite deterministic construction — a bijection (S, φ) on a finite configuration space — to observed physics. The reconstruction theorem addresses the converse at a narrower level: it asks which **local propagating lattice/gauge residue** the observed inputs fix, and answers that this residue is unique under the stated conditions. It does not select a unique Bell-inclusive full construction.

Stated as faithful synthesis of [Substratum]: given E1–E7 and A1–A6 together with M1-T, the C2 mixing hypothesis and the nearest-neighbor premise, the **local propagating lattice/gauge residue** is uniquely fixed. M1-B is separate. The reference nearest-neighbor graph is Bell-local under measurement independence for wings separated by more than its graph cone; the adopted branch uses preparation-indexed adjacency in the state-dependent graph $G(x)$ (or an equivalent ontically nonlocal response). H-Bell is the open requirement that those prepared graphs preserve operational no-signaling and the metric/Ollivier–Ricci continuum limit. That condition is downstream of a harder one. The cited convergence theorem is arbitrary-dimensional only for the manifold-distance-weighted graph metric, and for the degree-6 cubic reference graph the intrinsic hop metric is exactly ℓ_1 — a $\sqrt{2}$ diagonal stretch that does not decay at any scale — so the hop-metric curvature route fails for the reference family, before any Bell edge is added, and no $D = 3$ extension of that theorem would rescue it. The emergent geometry itself is not in question: the substratum's dispersion relation is isotropic at leading order. What must be re-founded is the curvature functional, on the propagation/Laplacian geometry rather than on shortest-path counts [SM §3.2]. No theorem here establishes Bell-inclusive full-substratum uniqueness. The reconstruction yields, as forced outputs rather

than fitted parameters, a finite set with a bijection of bounded coupling degree and statistical isotropy; spatial dimension three; the block multiplicities $(3, 2, 1)$; the Standard Model gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ (conditional on H-link, [SM §4.5]); three generations (physical reading conditional on H-spin', [SM §4.7]; chirality on H-χ', [SM §4.8]); one Higgs doublet (taken as input and accommodated in the E projection of the singlet taste, [SM §4.7, Theorem 11]); anomaly-free hypercharges (given the chiral structure and family pattern, [SM §4.9, Theorem 15]); and the relation $\hbar = c^3\epsilon^2/(4G\sigma)$, conditional on H-slope ([GR §3.2]), tying Planck's constant to the discretization scale, the gravitational constant and the entropy σ per boundary mode.

Two features of this theorem matter methodologically, independently of whether each forced output survives empirical scrutiny.

The first is the *direction* of the result. The theorem does not merely say that the construction can reproduce observed physics; it says the inverse inputs *back out* to a unique **local structural residue**. That is genuine pinning, but its domain matters: preparation-indexed Bell-composite data are not pinned by the theorem. The methodological content is therefore a demonstrated local structural residue plus an explicitly named completion underdetermination, rather than an unqualified unique ontology.

The second is the theorem's *honesty about its inputs*. The structural assumptions A1–A6 are not claimed as derived; they are restrictions on the class of constructions the reconstruction considers, and the framework states this. The uniqueness is uniqueness *within* the A1–A6 class with nearest-neighbor dynamics. This is not a concession that weakens the result so much as a specification of its scope: the theorem says that, holding fixed a small set of explicitly stated structural assumptions, the empirical inputs determine the structure. A reader who rejects an assumption in A1–A6 is not shown a contradiction; they are shown which alternative they must develop. The methodological discipline — every non-derived commitment named and exposed — is the same discipline Part I exhibited at the foundational layer, here operating at the level of the reconstruction.

There is also a scope limit the framework states itself, and faithful synthesis requires repeating it. The empirical inputs E1–E7 are idealized structural facts; no finite body of data establishes them outright. The theorem proves **local-residue uniqueness** under those idealized inputs. Reconstruction from finite data is a distinct and weaker question, answered operationally by the cumulative quantitative record rather than the theorem. This finite-data qualification is independent of the Bell qualification: even at the idealized-input level, Bell-inclusive completion uniqueness is open.

11. THE GAUGE-CLASSIFICATION THEOREM: EXHIBITING THE SURPLUS

A scoped reconstruction theorem establishes local-residue uniqueness up to *something* — up to the transformations that leave that reconstructed content invariant. An OSR claim needs that "something" itself to be exhibited and bounded. Otherwise the surplus is a residual unknown, and "the structure is real, up to we-know-not-what" is not yet a constructive position. The framework's second theorem supplies part of this: it exhibits generators of the gauge group $\mathcal{G}_{\text{sub}}$ — the transformations relating distinct constructions that produce identical observables — as an explicit list, whose exhaustiveness is open.

As faithful synthesis: the substratum gauge group $\mathcal{G}_{\text{sub}}$ contains three families of transformations; Theorem 24 adds a structural comparison of coupling graphs. *State relabeling* — conjugating the bijection by a permutation of the hidden states applied uniformly in every visible fibre, which preserves every channel (a fibre-dependent hidden relabeling preserves the transition probabilities but not coherent channels in general); this family alone is larger than any Standard Model gauge group. *Alphabet change* — replacing the local state space $\mathbb{Z}/q\mathbb{Z}$ by $\mathbb{Z}/q'\mathbb{Z}$ for any $q, q' \geq 2$, a generator under the alphabet-size hypothesis of [SM §2.7]: some results are proved for every alphabet, others carry q, and independence of every prediction from the choice is not proved. *Deep-sector enlargement* — adjoining a finite, independently prepared factor of hidden degrees of freedom beyond the boundary layer that no dynamics couples to it leaves every transition probability, and so the emergent description, unchanged; an extension coupled to the boundary layer with slow dynamics preserves only the leading-order transition law on accessible timescales and is not a generator. *Graph isomorphism up to statistical isotropy* — two coupling graphs with the same polynomial growth dimension, spectral mode count, and large-scale isotropy each admit a finite-law representation and share the growth exponent that enters the dimensional filters [SM §3.2], so in that sense the regular cubic lattice and a suitable bounded-degree random graph are structurally comparable; they need not give the same transition probabilities, and the continuum-curvature step [SM §3.1] has its own conditions, so this comparison is not a family of $\mathcal{G}_{\text{sub}}$. The Standard Model gauge group is the qualification: it depends on the octahedral point group of the cubic lattice (the (3,2,1) multiplicities), which a generic isotropic graph does not supply, so there the comparison is restricted to cubic-structure-preserving isomorphisms and the simple-cubic Bravais type is an empirically-selected input (see [Substratum §4, Theorem 24(iv)]).

The decisive word is *exhaustive*, and exhaustiveness is open. For the state-channel generators (state relabeling and deep-sector enlargement) the uniqueness clause of Stinespring's dilation theorem relates two dilations of the same visible-sector channel by a hidden-sector transformation, but equality of the full visible channel family does not extend this to a single conjugation of the generated dynamics (Substratum §4), so the classification's completeness is open. The graph comparison (iv) acts on the coupling-graph structure that feeds the

gauge derivation downstream of the visible-sector channel, establishes only a shared finite-law representation and growth exponent, with the cubic-structure qualification of [Substratum §4, Theorem 24(iv)] for the gauge sector, and is not a family of $\mathcal{G}_{\text{sub}}$. The surplus is exhibited; whether it is closed is open.

That a gauge freedom can be exhibited is worth contrasting with a known difficulty in the foundations of quantum field theory. In QFT with infinitely many degrees of freedom, the existence of unitarily inequivalent representations means that the choice of representation is a substantive interpretive problem rather than a settled one — a problem Ruetsche's analysis of interpreting quantum theories develops at length [Ruetsche]. The framework's setting is finite, and the gauge-classification theorem is what the finiteness buys: the analogue of "which representation" is "which representative within the equivalence class," and that question is not left as an open interpretive choice but posed as a finite classification, whose completeness is open. This is not a criticism of the QFT situation, which concerns genuinely infinite systems; it is a statement of what the framework's finiteness makes available — a finite classification problem, where the infinite case admits no analogous finite classification.

This is what converts the structural-realist claim from programmatic to constructive. The programmatic version says: there is a structure, and object-level descriptions carry surplus. The framework's version says: *here* is the structure — the equivalence class fixed by the reconstruction theorem — and *here*, in three enumerated families — whose completeness is open — is the exhibited surplus. The two theorems together leave, in the position's promissory note, exactly one named item: that completeness question. What is real is the equivalence class; what is gauge includes the three-family freedom; and which specific construction within the class one's universe "is" is — provably — without empirical content, because the gauge group relates all representatives to identical observables.

12. WHAT THE CONJUNCTION ESTABLISHES, AND WHAT IT DOES NOT

The methodological yield of Parts II is the conjunction, so it is worth stating exactly.

With both theorems in hand, the framework can do more than programmatic OSR at the **local reconstructed level**. It can exhibit the local residue, exhibit its gauge surplus through the three-family generators (whose completeness is open), and state exactly where the result stops. The local residue is not merely one economical option within the theorem's class; it is the residue the stated inputs fix. But realism about a uniquely specified Bell-complete substratum would outrun the theorem, because H-Bell does not classify or uniquely select preparation-indexed Bell-composite structure.

Three honest limits. First, the theorems are summarized here, not proved; their proofs and their dependency structure live in the framework's substratum paper, and the substratum paper itself flags that the reconstruction theorem's

confidence is bounded by the confidence of its dependencies. Part II inherits that flag and does not launder it. Second, the constructive-OSR claim is conditional on the structural assumptions A1–A6 and the nearest-neighbor premise, which are not themselves derived; the claim is "constructive OSR within the nearest-neighbor A1–A6 class," and the qualifier is not optional. Third, the claim is about the framework's particular construction. That a constructive OSR is achievable *here* does not show it is achievable for physics in general — but it does show the programmatic position has at least one constructive instance, which is itself a contribution to the OSR literature: an existence proof that the promissory note can be paid.

It is worth noting that the *idea* of reconstructing structure from observables is not foreign to mathematical physics. Algebraic quantum field theory contains a reconstruction result of its own — the Doplicher–Roberts theorem, by which the field algebra and the global gauge group are recovered from the category of physical representations of the observable algebra [Halvorson]. The framework's reconstruction theorem is not that theorem and does not depend on it, but the precedent matters methodologically: it shows that "the observable content determines the structure, including a gauge group, up to a specified equivalence" is a known *form* of result in rigorous physics, not a novel species of claim. What the framework adds, relative to that precedent, is the scope — a reconstruction not of a field algebra's gauge group but of a local substratum-equivalence-class from which the emergent quantum description, the Standard Model gauge group, and the gravitational sector all follow — and, with it, the exposure to a correspondingly larger set of structural assumptions, which is why the A1–A6 qualifier is load-bearing.

The methodological moral, which Part IV will generalize, is already visible. Once the structure is pinned and the surplus is classified, the question "but which substratum is the real one?" stops being a deep unanswered question and becomes a question shown to have no content. That dissolution — not the answering of the question but the demonstration that it was ill-formed — is the pattern Part IV generalizes.

PART III — PER-THEOREM CHECKS OF NO-GO RESULTS

Part III develops the framework's third methodological commitment: a common substratum, reduced description and trace-out architecture organizes separate checks of the no-go theorems' premises. The Bell and Wigner's-friend discussions are developed in Main; the Kochen–Specker and reality-theorem discussions here retain their stated conditional scope. The architecture locates the obligations, while each case requires its own supporting argument.

13. THE NO-GO THEOREMS AS A FAMILY

A deterministic-substratum account of quantum mechanics must answer to the no-go theorems of quantum foundations. These results — Bell's theorem [Bell], the Kochen–Specker theorem [KochenSpecker], the reality theorems in the Pusey–Barrett–Rudolph family [PBR], the Frauchiger–Renner theorem [FrauchigerRenner] — each show that some intuitively natural class of accounts of quantum mechanics cannot reproduce its predictions. A framework that posits a deterministic substratum underneath quantum mechanics is, on its face, exactly the kind of account these theorems constrain, and a reader encountering such a framework is right to ask, immediately, how it survives them.

The finite observable-law correspondence admits both a fixed-basis quantum representation and a deterministic realization. A local deterministic completion satisfying the setting and readout hypotheses of Main §3.3 nevertheless obeys $|S_{\mathrm{CHSH}}| \leq 2$. Within the assumed matrix and tensor setting of GR §3.3, equality with the quantum instruments instead requires the five completion conditions. The representation of a law and the operational premises of these theorems are separate mathematical requirements.

Bell constrains local response factorizability and the setting ensemble; Kochen–Specker constrains valuations of a noncommuting observable algebra; PBR constrains preparation independence of ontic composites; Frauchiger–Renner constrains the joint use of observer-relative predictions. The framework's hidden deterministic substratum and reduced observable law, related by the trace-out, provide a way to state these different questions. The fixed-basis quantum representation of the reduced law does not by itself settle their operational or composite premises.

The trace-out separates the two levels, but the four no-go theorems constrain distinct premises and require separate checks. Bell is sharper: for a response-complete deterministic completion, one-step own-wing read/write setting interventions, readouts outside the other wing's graph cone recording a ± 1 outcome at each wing in every run, and a setting-independent pre-setting ensemble give a Bell-local model and $|S_{\mathrm{CHSH}}| \leq 2$. The framework therefore cannot use the trace-out, screening, or indivisibility as an escape from Bell; it retains measurement independence and takes ontic parameter dependence for Bell-violating substrata, with operational no-signaling kept separately. Kochen–Specker constrains the noncommuting operational observable algebra, so its inheritance is conditional on the coherent operational lift. PBR depends on a preparation-independence postulate for the underlying variables; OI does not posit that postulate, but graph causal cones alone do not prove its failure or a particular quantum tensor factorization. Frauchiger–Renner turns on the relation between observer-relative operational descriptions and the substratum. What the four cases genuinely share is the need to keep substratum statements, fixed-basis representation statements, and full operational-QM statements at their correct levels. The trace-out is the common architecture, not a universal theorem-

disabling device.

14. THE FOUR CASES

Bell's two layers. Spatial OI gives an exact graph causal cone for finite-range substrata. But when the completion is response-complete and deterministic, one-step own-wing read/write setting interventions, readouts outside the other wing's graph cone recording ± 1 at each wing in every run, and one measurement-independent pre-setting ensemble give a Bell-local hidden-variable model and the classical ceiling $|S_{\mathrm{CHSH}}| \leq 2$. The framework therefore does not use screening plus graph locality as its route to quantum Bell correlations: it declines the measurement-dependent/superdeterministic branch and takes ontic parameter dependence for Bell-violating substrata, with operational no-signaling required. The Barandes–Hasan–Kagan Tsirelson result remains a statement about the weaker stochastic causal-local indivisible description, not about the deterministic completion. Kochen–Specker is conditional on the common coherent non-commuting instrument lift. The PBR discussion is likewise conditional at the composite level: OI does not postulate unconditional preparation independence, and co-embedding permits shared hidden data, but the exact composite operational structure is not supplied by the graph cone alone. Frauchiger–Renner receives an observer-relative account from partition-relative reduced descriptions, with its full-protocol premises assessed separately. These different statuses are kept separate below.

Bell's theorem. Bell's theorem binds on the framework because the substratum configuration is a hidden variable. The dependency graph proves a finite-propagation statement for any substratum whose Bell-relevant couplings are finite-range: until the remote setting's cone reaches the local readout region, the remote setting cannot change the local response through that graph. If the setting interventions are applied at one step, each reading and writing only its own wing, each wing records a ± 1 outcome in every run outside the other wing's graph cone, and a pre-setting variable fixes both wings' responses to every setting with an ensemble independent of the settings, determinism turns the model into an ordinary local hidden-variable decomposition, hence $|S_{\mathrm{CHSH}}| \leq 2$. That is the Bell ceiling theorem of Main §3.3. Quantum Bell violation therefore cannot be attributed simultaneously to that pointwise cone and measurement independence.

The framework's chosen branch is ontic parameter dependence: one wing's response can depend on the remote setting at the microstate level, with operational no-signaling required. Whether preparation-indexed adjacency supplies that dependence while preserving no-signaling and the continuum-curvature limit remains the open H-Bell compatibility question. The alternative is ontic measurement dependence; Main records but does not adopt it, with the necessary bound $S \leq \min\{4, 2+4\sqrt{2 \ln 2\, I_{\mathrm{ont}}}\}$. The stochastic causal-local indivisible construction can obey Tsirelson because its causal-independence condition is weaker than Bell factorization; that result is retained at the stochastic layer only. Un-

restricted globally controlled OI realizations can produce PR-box behavior, so no-signaling, hidden memory, and indivisibility are all insufficient by themselves to select the quantum correlation set. *(Established: graph causal-cone theorem for finite-range substrata, Bell ceiling for the screened deterministic completion, and the measurement-dependence bound; adopted Bell branch: ontic parameter dependence with operational no-signaling.)*

The Frauchiger–Renner theorem. For its specified multi-agent protocol, Frauchiger–Renner [FrauchigerRenner] derives a conflict among quantum probability-one certainty, consistency of agents' certainty inferences, and single outcomes, denoted (Q), (C) and (S). Distinct state assignments arising from different observations in the simpler Wigner's-friend scenario do not by themselves exclude universally valid quantum mechanics. Main's partition construction supplies observer-relative marginal laws and a joint law with those marginals; it does not establish that a particular Frauchiger–Renner premise fails.

Applying that theorem to a proposed OI completion requires the operational probabilities of its full protocol and an analysis of the probability-one, consistency and single-outcome premises. Definite substratum configurations and partition-relative reduced descriptions alone do not decide those premises. The Wigner's-friend construction is an observer-relative interpretation of a common underlying law; an unconditional Frauchiger–Renner resolution does not follow from it.

The Kochen–Specker theorem. Kochen–Specker shows that no noncontextual hidden-variable assignment — one assigning every observable a definite value independent of which compatible set of observables is measured alongside it — can reproduce quantum mechanics. The framework's relation to it is a short *conditional* argument on results the framework already has. The framework reproduces the quantum statistics at the fixed-basis transition layer: the characterization theorem delivers a full equivalence there ($S \iff D \iff Q_{\mathrm{fb}}$, Main §3.4). Kochen–Specker is a theorem *about* the operational theory — it states that the observable algebra of quantum theory in dimension three or higher admits no noncontextual value assignment — and that algebra of noncommuting observables belongs to the operational bridge, which Main characterizes conditionally, with its physical sourcing open (§3.2, §3.4). The inheritance is therefore conditional and automatic: if the coherent operational lift reconstructs the standard observable algebra, any account carrying it carries a structure with no noncontextual valuation, and the framework's emergent valuation is necessarily contextual, on pain of contradicting that operational structure. Contextuality is thus not a property the framework must be argued into case by case — it is forced by the equivalence, once the bridge is crossed.

What the framework adds is the *account* of that contextuality, and here the Section 13 architecture does the work. An emergent observable's value is produced by the trace-out, and the trace-out is taken with respect to a partition; measuring a compatible set of observables is a specific apparatus coupling, and

the framework models a measurement context as exactly such a coupling-and-partition configuration (compare the double-slit case of Main §4.1, where coupling a which-path record is an intervention that changes the visible transition law). Which emergent observables are jointly defined, and the value each takes, are features of the reduced description fixed by that partition — not partition-independent functions of the substratum configuration. Contextuality, in the framework, *is* the partition-dependence of emergent valuation: the same partition-relativity Part II identified as gauge. The framework need not reconstruct the Kochen–Specker colouring obstruction itself; that obstruction is inherited intact with the operational theory, once the bridge selects it. The framework is a deterministic hidden-variable theory whose hidden level is a definite substratum configuration but whose emergent valuation is contextual — precisely the kind of hidden-variable theory Kochen–Specker leaves open. *(Conditional: this paper, on the Main paper's equivalence theorem together with the coherent operational lift that Main characterizes conditionally, §3.2, §3.4.)*

The reality theorems (Pusey–Barrett–Rudolph family). The PBR theorem shows that, given a natural assumption about how independently prepared systems combine — the Preparation Independence Postulate, that the joint ontic-state distribution of two separately prepared systems is the product of the single-system distributions — the quantum state cannot be a mere probability distribution over deeper states of reality, but must be ontic. A deterministic-substratum framework is naturally read as making the quantum state epistemic: the state is the observer's representation of incomplete knowledge of the substratum configuration. The framework's relation to PBR turns on the Preparation Independence premise and on the still-separate composite operational lift; it is therefore conditional rather than already closed by the fixed-basis equivalence.

The Preparation Independence Postulate requires an unconditional product distribution for the ontic states of independently prepared systems. OI does not impose that postulate at the substratum level: two preparation devices embedded in one substratum may share preparation-relevant hidden or boundary data even when no later dynamical signal passes between their laboratories. This is enough to show that PBR cannot simply be imported as a theorem about bare OI without first establishing its preparation-independence premise. The point should not be strengthened into a theorem that graph locality itself yields a particular conditional product distribution or a quantum tensor-product state: the graph gives causal cones, while the exact composite/tensor operational structure belongs to the coherent operational lift still stated separately in Main. Consequently the framework's relationship to PBR remains conditional at the composite layer. If a proposed OI completion were shown to satisfy PBR preparation independence together with the other hypotheses, it would inherit PBR's conclusion; if it does not, the theorem is outside that completion's scope.

15. WHAT THE COMMON ARCHITECTURE CONSTRAINS

The common architecture has mathematical content without deciding all the no-go premises. A finite bijection fixes one configuration trajectory from each initial state, and its finite observable law admits a fixed-basis quantum representation. That representation theorem is universal, including for Markov laws, and does not supply every operational quantum preparation, instrument or composite [Main §3.4].

The four cases are not four versions of one theorem-disabling trick. They share one bookkeeping architecture — substratum, reduced description, and trace-out — but each no-go theorem constrains a different object and must be checked separately. Bell is the clearest warning against over-unification: Bell factorizability fails for the observed quantum correlations themselves, and a response-complete deterministic completion cannot recover those correlations from graph locality plus measurement independence. The adopted Bell-violating completion is therefore ontically parameter-dependent, with operational no-signaling required. Frauchiger–Renner concerns the joint use of observer-relative predictions under its certainty, consistency and single-outcome premises; PBR concerns preparation independence at the ontic composite level; Kochen–Specker concerns noncontextual valuations in the noncommuting operational algebra. These are distinct restrictions on specified substratum or operational objects; the trace-out alone does not establish that the relevant premise fails.

Two results show the distinction. The deterministic Bell ceiling gives $|S_{\mathrm{CHSH}}| \leq 2$ under the stated local-response and setting-independence premises, even though the observable law has a fixed-basis quantum representation [Main §3.3]. The operational-completion characterization, within its assumed complex matrix kinematics and tensor composite, identifies the available quantum instruments exactly under five additional conditions that bare OI does not imply [GR §3.3]. Thus the shared representation constrains the objects in the argument while leaving substantive operational premises to be supplied.

The Bell and Frauchiger–Renner discussions are developed in Main and summarized here; the Kochen–Specker and PBR discussions in Section 14 remain conditional on their separately stated premises. Fixed-basis representation equivalence alone supplies neither the noncommuting operational algebra nor the quantum tensor-product composite. The operational-completion characterization assumes complex matrix kinematics and characterizes available instruments under its five completion conditions; the derivation of that kinematics and the physical sourcing of those conditions remain open. The common architecture organizes these distinct constraints without discharging them by itself.

PART IV — PHYSICS MODULO GAUGE

Part IV draws the three preceding commitments into a single methodological proposal: that the well-posed question for fundamental physics is not "what is fundamental?" but "what is the equivalence class of structures that produce the observations, and what are its representatives?" This part argues the proposal as a general methodology, not as a feature peculiar to the framework's particular construction.

16. THE QUESTION THAT DOES NOT CLOSE

The realism debate in the philosophy of physics is organized around a question: what is fundamental? Realists answer with an inventory — particles, fields, spacetime points, a wavefunction on configuration space — and divide over which inventory is correct. Anti-realists decline the inventory and divide over what to say instead. The debate is durable, and its durability is itself a datum. A question that has resisted resolution for as long as this one has — through every major change in physical theory, with the candidate inventories turning over each time while the debate's structure persists — invites the suspicion that the question is not hard but ill-formed.

The three preceding parts suggest where the ill-formedness lies. Part I found that "observation occurs," the framework's apparent single foundation, was not one claim but two, and that stating it at the right granularity meant resolving it into an indubitable axiom and a separable posit. Part II found that the structure consistent with the empirical inputs has a unique local residue — and that the gauge freedom is not noise to be discarded but a group with three exhibited families, whose exhaustiveness is open, specifying differences between descriptions that make no difference. Part III distinguishes the local-response, operational-algebra, preparation and agent-consistency premises of the no-go arguments; the shared substratum and observable objects do not establish that any particular premise fails.

The common thread of Parts I and II is that each apparent puzzle was a failure to track a distinction the framework's structure makes precise: between what is committed and what is evidential, and between what is structural and what is gauge. "What is fundamental?" fails to track the first of these in the same way. It presupposes that among the structures consistent with the evidence there is a fact of the matter as to which is *the* real one — that the gauge freedom Part II exhibits is not gauge after all but conceals a genuine further fact. The reconstruction and gauge-classification theorems say there is no such further fact: within the equivalence class the choice of representative is, provably, without empirical content. "Which representative is the real one?" is not a hard question with an unknown answer. It is a question the framework's own structure certifies as empty.

17. THE WELL-POSED REPLACEMENT

If "what is fundamental?" is ill-posed, it should be replaced, not merely abandoned. The replacement the framework's methodology proposes is: *what is the equivalence class of structures that produce the observations, and what are its representatives?*

For the reconstructed local sector this question is well-posed in a determinate form: the reconstruction theorem gives the local equivalence class, and the gauge-classification theorem gives its representatives by enumerating the relevant transformations, whose completeness is open. The Bell-inclusive completion adds a third question — which compatible preparation-indexed Bell rule, if any — whose uniqueness is presently open. The question decomposes into two sub-questions each of which is answerable by a theorem rather than by a stance: a uniqueness question (what class?) and a classification question (what gauge freedom within it?). Where "what is fundamental?" asks for a winner among the representatives and the framework certifies that there is none to be had, "what is the equivalence class and what are its representatives?" asks for exactly the two things the framework's theorems address.

Call this posture *physics modulo gauge*. The name is chosen to make the structure explicit. Physics, on this view, determines its content up to gauge — up to a classified equivalence — and the determination *up to gauge* is not a limitation to be apologized for but the complete and correct form of the answer. The realist's mistake is to want physics modulo nothing, a single representative crowned; the anti-realist's mistake is to conclude from the absence of a crowned representative that physics delivers no structural reality at all. Physics modulo gauge is the position that the equivalence class is real, fully and determinately, and that the gauge freedom is real too — real *as gauge*, that is, as the exactly specified set of distinctions that carry no empirical content. Neither the class nor the gauge freedom is a defect. Together they are the answer.

The posture has affinities with existing positions, and locating it among them sharpens it. Healey's analysis of gauge theories [Healey] argues that gauge freedom is not mere descriptive redundancy to be quotiented away without loss — that there is interpretive content in how a gauge theory's surplus structure is handled; physics modulo gauge agrees that the gauge freedom is not nothing, and adds that, when the gauge group is *exhaustively classified* — which Part II leaves open — the freedom becomes a determinate object of knowledge rather than an interpretive residue. Wallace's "mathematics-first" formulation of structural realism [Wallace2022] — on which one states the physics in its mathematical form and reads the ontology off that, rather than first fixing an object ontology — is the variant of structural realism closest to the present posture; physics modulo gauge can be seen as a mathematics-first structural realism in which the "mathematics" is specifically the equivalence class fixed by the reconstruction theorem and the gauge group classifying its representatives. The difference from the programmatic versions, as Part II stressed, is

constructiveness: the class and the group are exhibited, not promised.

The posture reframes several standing puzzles, beyond the realism debate itself. The "unreasonable effectiveness of mathematics" becomes, in this setting, a scoped reconstruction result rather than a mystery: the observed inputs and the **local reconstructed structural residue** determine each other under the theorem's assumptions. Bell-composite structure is not included in that bidirectional uniqueness claim, so the residual underdetermination includes both the stated structural assumptions and the open Bell-inclusive completion. The question whether mathematics *is* reality or merely *describes* it is, *for the embedded physical observer*, exhibited as gauge — the two phrasings are related by a transformation that changes no observable an embedded observer can perform, so the question of which is correct has, for physics, the same status as the question of which substratum representative is the real one. This is gauge *in the framework's specific physics-internal sense*: the framework's gauge classification is relative to what an embedded observer can measure, not relative to all possible vantage points. The framework does not claim the question is absolutely meaningless. Whether the question has substantive content in mathematical foundations, meta-mathematics, or other modes of inquiry not bound by embedded physical observation is a separate matter the framework does not address; what the framework establishes is that the question carries no empirical weight for physics. A specialist in mathematical foundations is free to pursue it; the framework's reconstruction theorem provides, modulo gauge, the specific mathematical structure that would be in question if it has a positive answer, without committing to whether it does. The measurement problem, on this posture, receives an observer-relative account rather than an added mechanism: at the represented fixed-basis level, measurement conditions on a visible outcome and re-marginalizes the hidden sector, the definiteness of outcomes is a feature of every partition-relative emergent description, and the apparent conflict between descriptions is the partition-gauge covariance of Part III ([Main §3.4, §4.1]). Multi-agent protocols, the Frauchiger–Renner premises among them, require the operational completion's probabilities and a separate premise check (Section 14).

A further structural observation, kept brief here because it lies outside the framework's primary scope: if a mathematical-foundations specialist pursues the question and adopts a Tegmark-style position that all mathematical structures have ontological status, the framework's reconstruction theorem provides a natural *selection principle* within that class. The structures that admit embedded observers satisfying the framework's structural conditions (C1–C4 of the framework's Main paper, plus the gauge classification developed in Part II) form a precisely characterizable sub-class — the "observer-supporting" structures. This addresses what is sometimes called the measure problem for mathematical-universe ontologies: given that all mathematical structures are taken to be real, why does our universe instantiate the specific structure it does? The framework's answer, available to anyone pursuing the question in that direction without requiring commitment to the underlying ontology: the structures we can

observe are precisely the observer-supporting ones, characterized as a selection effect. The framework does not commit to a mathematical-universe ontology; it provides the selection principle that any such ontology would benefit from having if it is pursued.

18. THE PROPOSAL AS A GENERAL METHODOLOGY

The claim of this part is not merely that the Incompleteness framework happens to admit a physics-modulo-gauge reading. It is that physics modulo gauge is a methodology — a way of posing and answering foundational questions — that other programs in fundamental physics could adopt, and against which they can be measured.

The proposal can be located against a long-standing open problem. Hilbert's sixth problem [Hilbert] called for the axiomatization of physics — the treatment of physics, in those parts ready for it, with the rigour of an axiomatic mathematical theory. More than a century on, the problem is generally regarded as only partially addressed, and a recognized obstacle is that the primitive notions of physical theories — "body," "force," and their successors — tend to be, in the phrase of the problem's modern commentators, pregnant with theory rather than genuinely primitive. The methodology proposed here bears on that obstacle directly. Its discipline is to identify the genuine primitives by stripping the theory-laden ones away — Part I's distillation of "observation occurs" to a two-axiom base is exactly this move, performed at the foundational layer — and then to ask, of the structure those primitives plus the empirical inputs determine, the two well-posed questions of physics modulo gauge. The proposal is, in this sense, a partial and specific response to Hilbert's sixth problem: not an axiomatization of physics entire, but a method for stating what a particular foundational program commits to, with the primitives exposed and the gauge freedom classified.

Stated generally, the methodology has three steps. First, for a candidate fundamental theory, identify the empirical inputs and ask whether they determine a structure uniquely — whether a reconstruction result is available. Second, where uniqueness holds only up to some equivalence, classify the equivalence explicitly: exhibit the gauge group, and prove the classification exhaustive, so that the surplus is bounded and known. Third, having the class and its gauge group, decline the question of which representative is real, and certify that declension with the gauge classification itself — the representatives are inter-transformable by observable-preserving maps, so the question has no empirical content.

A program that carries out these three steps has, on this view, said everything there is to say about what its theory commits to. A program that cannot carry out step one — that cannot exhibit its structure as determined by the evidence — has not yet earned a realist reading of that structure, whatever its other merits. A program that carries out step one but not step two — that has a structure but

no classification of its gauge freedom — has an unbounded surplus and cannot yet say which of its commitments are genuine. The methodology is, in this sense, a standard: it specifies what a foundational program must exhibit before its ontological claims are well-posed, and it does so in a way that is checkable, because each step is the kind of thing that is done by a theorem or not done at all.

This is the sense in which physics modulo gauge is offered as a contribution beyond the framework. The framework is one worked example — a case where the three steps have been carried out up to the open exhaustiveness of step two's classification, on which step three's certification also rests, with the reconstruction theorem for step one, the gauge-classification theorem for step two, and the certified declension for step three. But the steps are statable without reference to the framework's particular substratum, and a different program with a different ontology could carry them out differently. What the methodology asserts is that *until* they are carried out, in some form, the question "what is fundamental?" remains ill-posed for that program — and that once they are carried out, the question has been replaced, not answered, by the two well-posed questions that physics modulo gauge substitutes for it.

The honest scope of the claim: that physics modulo gauge is the *correct* general methodology is not established by one worked example, and Part IV does not claim it is. What the framework establishes is weaker and still substantive — that the methodology is *instantiable*, that there exists at least one foundational program in which the three steps are carried out, up to that open exhaustiveness, and the realism question is thereby reframed rather than left open. Whether the methodology generalizes — whether other foundational programs can or should be held to the same three steps — is itself left as the proposal's open horizon, in the same spirit in which Part I left the succession no-go open to a derivation that has not been found. The methodology is offered for use and for criticism, not asserted as proven.

19. GAUGE FREEDOM AND OBSERVATIONAL INCOMPLETENESS

This section is the most recent addition to the paper and the least settled; it is developmental even by the standards of the rest, and is offered as a proposal for scrutiny. It connects the methodology of Parts II–IV to the framework's core physical thesis — the incompleteness of observation — and argues the two are, on a precisely delimited class, one principle.

The framework takes its name from a thesis in physics: that an observer, being a proper part of the system it observes, cannot fully resolve that system — the trace-out over the unobserved remainder destroys information that no embedded observer can recover. Observational incompleteness, so stated, sounds like a limitation: a deficit, a wall past which the observer cannot see. Parts II through IV have been about something that sounds quite different — gauge freedom, the

surplus in a description, the respects in which distinct representations say the same thing. A deficit and a surplus appear to be opposites. This section argues they are not — that on a precisely identified class of cases, gauge freedom and observational incompleteness are the same fact seen from two sides, and that recognizing this is what lets the methodology of physics modulo gauge connect to the framework's foundational physics rather than sit beside it.

Begin with the apparent opposition, because dissolving it is the crux. Gauge freedom is descriptive redundancy: two representations, no observable difference, the surplus discardable. Observational incompleteness is missing information: a real system, an embedded observer, a fact the observer cannot reach. Surplus versus deficit. But ask *why* a gauge difference is surplus. The state-relabelings of the framework's substratum gauge group are surplus precisely because no observation distinguishes the relabeled substrata. The redundancy is not a free-standing fact about the representation; it is *constituted by* the observer's inability to tell the representatives apart. "There is surplus structure here" and "the embedded observer cannot access this difference" are not opposed claims — they are one situation, described from the representation side and from the observer side. The surplus *is* an inaccessibility; the inaccessibility *is* what makes it surplus.

This identification must be delimited, because not every observational incompleteness is gauge, and the cases that are not are as instructive as the cases that are. Consider two ways the framework's embedded observer fails to have a fact.

In the first, there is no fact to have. The extent of a decoupled hidden enlargement, the labels on hidden substratum states, the question of whether the underlying contrast is tenseless or temporal (the subject of this paper's foundational Part I) — for each of these, the framework's own gauge classification establishes that *no observable depends on the answer*. There is not a hidden truth the observer is failing to reach; there is no truth there to reach, because a difference that no observation could ever register is not a difference in fact. Call this **gauge incompleteness**: the observer's picture is incomplete in the bookkeeping sense that a question can be posed, but the incompleteness is empty, because the question has no answer-bearing content. This species of incompleteness *is* gauge freedom — the name records the identification, which is a result and not a stipulation.

In the second, there is a definite fact, and the observer still cannot reach it. The observer's own future state, under the deterministic substratum dynamics, is fixed — there is a fact of the matter about it. But to compute that future the observer would need to be a system at least as large as the whole of which it is a proper part, and it is not; it is $V \subsetneq S$. Here the incompleteness is genuine and the missing item is a real fact, withheld not by the absence of content but by the observer's finite computational reach. Call this **computational incompleteness**. It is *not* gauge — a larger computer, still inside the same physics, could reach the fact; nothing about it is surplus.

The distinction between these two species is, as far as we are aware, not drawn in the existing literature on observational limitation, and drawing it is part of this section's contribution. With it in place, the identification can be stated exactly. *Gauge freedom is identical to gauge incompleteness — the species of observational incompleteness in which the inaccessible question has no answer-bearing content.* It is not identical to computational incompleteness. "Physics modulo gauge," the methodology, and "observational incompleteness," the physics, name one principle restricted to the gauge-incompleteness species: the methodology-side name and the physics-side name of the same fact.

One objection must be met head-on, because it is the natural one for a reader versed in the philosophy of gauge theories. It will be said that "surplus structure" is contested — that there is a body of work (Nguyen, Teh, and Wells [NguyenTehWells]) arguing surplus structure is not eliminable but essential, and that an identification of gauge with a *species of incompleteness* cannot be cleaner than the contested notion it leans on. The reply is that the contest is real but does not touch this identification, and the reason is a distinction drawn by Bradley and Weatherall [BradleyWeatherall]: "surplus structure" carries two nearly opposite senses. In one — *formulational* surplus — the structure is what a theory needs in order to be written as a local field theory; this is the sense in which surplus is essential, and the Nguyen–Teh–Wells result concerns it. In the other — *interpretational* surplus — the structure is descriptive redundancy with no fact of the matter as to which representative obtains. The framework's gauge — the substratum gauge group, classified by the gauge-classification theorem, with the equivalence class taken as what is real — is interpretational surplus, unambiguously: the theorem classifies transformations that preserve every observable. The "surplus is essential" result lives in the formulational sense and does not transfer. The identification of gauge with gauge incompleteness inherits no controversy it has not earned; it concerns interpretational surplus, and for that sense the "no fact of the matter" characterization is exactly standard.

There is a consequence of this identification that is worth stating in closing, because it is the point at which the paper's methodology turns back on the paper's own first part. If gauge freedom is the gauge-incompleteness species of observational incompleteness, then physics modulo gauge — the methodology — is a methodology *for working in the presence of observational incompleteness.* And observational incompleteness, in this framework, is not a special circumstance; it is the condition of any embedded observer, which is to say of any observer at all. The methodology is then not a special technique for a special class of theories. It is the general form that honest physics takes once it is admitted that the physicist is inside the system. Part I found that the framework's own foundation, examined closely, has a feature — whether the underlying contrast is tenseless or temporal — that is gauge: a question the framework can pose and that no observation it could ever make will answer. That the framework's *own foundational description* contains gauge incompleteness is not an embarrassment to be hidden. It is the methodology applying to itself, exactly as it should, if the methodology is correct: the physicist is inside the system even

when the system in question is the physicist's own foundation.

The gauge-incompleteness / computational-incompleteness distinction admits a wider reading worth flagging, since it connects this paper's methodology to a strand in recent epistemology. Deutsch [Deutsch] argues in *The Beginning of Infinity* that knowledge itself is structurally incomplete — that from any current state of explanatory understanding there is always more knowledge to be created, and this incompleteness is a constitutive feature of what knowledge is rather than a contingent limitation awaiting some final theory. The distinction the present section draws between two species of observational incompleteness aligns naturally with a distinction Deutsch's account requires though does not state in these terms: gauge incompleteness is *structural* — the question has no answer-bearing content because no observation could distinguish — and computational incompleteness is *contingent* — the answer exists but requires more capacity than the observer has. The structural form is the one the framework's gauge-classification theorem identifies; the contingent form is the one Deutsch's account is built around. Both are real species of incompleteness; only the first is gauge.

The connection has a direction. Deutsch's argument is independent of any specific physics — it argues from explanatory creation being open-ended — but the framework provides one physics-foundational route to grounding it. Knowledge derived from observation cannot reach what observation cannot reach; the gauge-classification theorem establishes that observation, structurally, cannot reach the substrate's gauge equivalence class; therefore knowledge built from observation must inherit observational incompleteness as its lower bound. The reverse implication does not hold automatically — BoI does not presuppose this framework's specific machinery. What the framework adds is a stronger and more specific grounding than BoI provides on its own: knowledge is incomplete not only because explanatory creation is open-ended, but also because observation, the source of empirical input, is itself structurally incomplete in a way the gauge-classification theorem makes precise. The methodology of this paper is therefore one specification of what Deutsch's epistemological commitment requires of any foundational program that takes observation seriously: identify the gauge incompleteness, exhibit it constructively, and decline the questions that fall inside it as ill-posed rather than unanswered.

PART V — THE OPERATIONAL METHODOLOGY

Part V develops the methodology of the preceding parts at the operational level: given the substratum gauge group classified by Part II, what does substrate-only computation determine, and what does it provably not determine? The operational answer turns out to be non-trivial in both directions, with two no-

go theorems and the systematic audit results developed in the technical papers serving as the concrete demonstrations. Part V is more recent than Parts I–IV and is offered as developmental in the same sense the earlier parts are — provisional, but with each section's specific developmental status flagged.

21. THE OPERATIONAL QUESTION

Parts II through IV established that physics modulo gauge is the right *form* of the answer for a foundational program with a constructively classified gauge group. The well-posed questions, on that account, are what the equivalence class produces and what its representatives are; the ill-posed question is which representative is real. What Parts II–IV did not yet ask, and what Part V develops, is the operational form of the same question: given that the equivalence class is fixed by the reconstruction theorem and the gauge group classified by the gauge-classification theorem, which quantities are determined by substrate-only computation, and which depend on the choice of representative inside the equivalence class?

The question is non-trivial in both directions. Some quantities the framework would *like* to derive from substrate computation alone turn out to depend on choice of representative; the operational methodology forces these to be classified differently than the framework's first-pass framing would suggest. Other quantities turn out to be substrate-derivable in ways that the framework's first-pass framing did not make explicit; the operational methodology surfaces these as content that had been undercounted. Part V develops both directions through two no-go theorems (§22, §23), the comparison with Part III's distinct premise checks (§24), the systematic audits the technical papers carry out (§25), and the methodological refinements the audits canonize (§26).

The status of Part V as a whole is the same as that of Parts II–IV: the methodology is a proposal, the worked instances are developed results in the framework's technical papers, and the unification stated here as synthesis is the present paper's contribution. Each section flags its own status more specifically.

22. THE GAUGE-INVARIANCE NO-GO

Status: developed result; the theorem and its proof are in the framework's technical record. The statement and proof sketch below are faithful to that source; the methodological reading is the present paper's synthesis.

The first operational result is a no-go theorem at the level of substrate computation. State it sharply. Let $\mathcal{G}_{\text{sub}}$ denote the substratum gauge group classified in Part II (the group of substratum transformations that preserve every observable accessible to an embedded observer). Let a *substrate-only computation* be any procedure that operates on a single representative of the substratum equivalence class and produces a numerical output. Let a *bijection-specific quantity* be any quantity whose value depends on which representative is chosen within

the equivalence class — equivalently, any quantity that is not $\mathcal{G}_{\text{sub}}$-invariant. The theorem states that no substrate-only computation determines a bijection-specific quantity.

The proof is short. By the gauge-classification theorem (Part II), the equivalence class $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$ is the framework's content; every observable depends only on equivalence-class invariants. A bijection-specific quantity, by definition, takes different values on representatives related by $\mathcal{G}_{\text{sub}}$. A substrate-only computation, by definition, operates on one representative. If the computation's output is a bijection-specific quantity, then the same computation applied to a $\mathcal{G}_{\text{sub}}$-related representative produces a different output. But no observable depends on the choice of representative. The computation therefore produces an output that is not an observable — it produces a value, but the value has no empirical content. Substrate-only computation, on this account, can determine an equivalence-class invariant or it can produce a non-observable; it cannot determine a bijection-specific quantity that is also empirically meaningful.

The reading of the theorem is the methodologically important part. A bijection-specific quantity is not a forbidden quantity; it is a quantity whose value is selected by the empirical inputs that pick the representative, not by the substrate's structure. Three quantities the framework's empirical record contains illustrate the point. The directional content of the framework's Condition 2 sum rule — which pair of mixing-angle ratios receives the 4/3 coefficient — depends on the assignment of mass eigenstates to generations, an assignment made by experimental measurement rather than by substrate computation. The CP-violating phases of the CKM and PMNS matrices are bijection-specific in the same sense: the absolute phases require fixing a flavor basis, and the basis is selected by which physical particle is identified with which substrate state. The mass-hierarchy assignment within a generation — which of the two states in a pair is the heavier — is bijection-specific in the same way. Each of these quantities the framework reproduces in its empirical record; each does so by taking the experimental assignment as input rather than by deriving it from substrate computation alone.

The methodological lesson is that the boundary between substrate-derived content and empirically-input content is sharper than a framework's first-pass framing typically makes it. A claim that "the framework derives the 4/3 sum rule" fails at both levels: at the magnitude level the coefficient $4/3 = 2A^2$ is an equivalence-class invariant of the cubic-group representation theory, but the substrate is not shown to produce it as the coefficient ratio (the second-order substrate route is excluded, [SM §7.3]); at the directional level which pair-ratio gets the 4/3 is bijection-specific, fixed by the empirical mass-hierarchy assignment. Both statements are about the same prediction; the methodological discipline is to keep them separated. This separation is what the operational methodology of Part V proposes as the standard for any foundational program of this shape: state explicitly, for each prediction, what is gauge-derivable and what depends on the empirical inputs that pick the representative.

The theorem has a converse that should be stated, since it is the part that prevents the result from being mistaken for a deflationary one. Quantities that *are* $\mathcal{G}_{\text{sub}}$-invariant — the equivalence-class invariants — are determined by substrate computation, and the framework's empirical record contains candidates for them: the Cabibbo magnitude $\lambda = 1/(\pi\sqrt{2})$ and the Koide ratio 2/3 (conditional structural, [SM §7.6]), the Bekenstein–Hawking entropy coefficient 1/4 (a consistency relation under H-slope, [GR §8.5]), the MOND acceleration scale $a_0 = cH/6$ (its dimensional factor imported, [GR §7.3]), and the structural forms of the mixing angles. Their magnitudes are not bijection-specific; what the substrate determines of each is stated entry by entry in [SM §7.6] and [GR §8.5]. The gauge-invariance no-go does not deflate the framework's substrate-derivable content; it draws the line between that content and the content that requires empirical input. Where the line falls is the operational question of §22; the answer is not vacuous in either direction.

23. THE CATEGORY-ERROR FINDING

Status: developed result; the type-mismatch is established by direct inspection of the framework's substratum-level structure (see the framework's substratum paper, §2.5, and the technical-paper record more broadly). The methodological reading is the present paper's synthesis.

The second operational result is more pointed than the first. It states that some questions the framework's first-pass framing makes available are not merely gauge-forbidden but ill-typed: the question asks for an object the framework's substratum-level structure does not contain.

The instance through which the finding was identified is the substrate-level Lagrangian. The framework's quantitative content (mass ratios, mixing angles, coupling thresholds) takes Lagrangian form at the *emergent* level: post-trace-out, after the substratum dynamics have been marginalized over the unobserved hidden sector, the visible-sector dynamics admit a description as a quantum field theory with an action functional. The substrate-level dynamics, by contrast, are *not* given by an action functional. The substrate is the deterministic linear bijection $\varphi : S \to S$ of the foundational definition; its dynamics is the iteration of φ, not the extremization of an action. A "substrate-level Lagrangian" is therefore not a thing the framework's substratum-level structure offers; the construction has no place to attach.

The mismatch is type-level rather than calculational. It is not that a substrate-level Lagrangian is hard to compute or that the relevant calculations are beyond current tools; it is that the request itself does not pick out a well-typed object in the framework's foundational ontology. Asking for "the substrate Lagrangian" is structurally similar to asking for "the temperature of a single classical particle" — temperature is a thermodynamic quantity defined on ensembles, and applying it to a single particle is not a difficult question but a category error. The framework's substrate is the bijection on a finite set; Lagrangian field theory

is the post-trace-out description that emerges from marginalizing the bijection over the hidden sector. The two live at different levels of description, joined by the trace-out, and the joining is one-directional: emergent Lagrangians come *from* the substrate by trace-out, but there is no inverse map that produces a substrate-level Lagrangian *from* the emergent one.

The methodological point generalizes beyond the Lagrangian case. Any framework with a level distinction — substrate and emergent, fundamental and effective, microscopic and macroscopic — will have constructions natural at one level that do not transfer to the other, and not all such non-transfers are blocked by gauge invariance in the §22 sense. Some are blocked by type-mismatch: the construction is not defined at the level it is being asked of. The methodological discipline is to recognize the difference. A gauge-forbidden quantity exists at the relevant level but cannot be picked out by substrate-only computation; a type-mismatched construction does not exist at that level at all. Conflating these blocks two different kinds of inquiry under the same label and obscures what the framework's substrate-level commitments actually are.

The category-error finding has a corrective consequence for the framework's own research program. The framework's technical record contains a sequence of attempts at substrate-level closure of certain quantitative predictions — attempts that constructed candidate substrate-level Lagrangians and looked for the framework's coefficients to fall out of those Lagrangians. The category-error finding establishes that such attempts were not failing for calculational reasons; the targets they aimed at were not well-typed objects in the framework's foundational ontology. Re-classifying the relevant quantities as either equivalence-class invariants (substrate-derivable in the sense of §22) or bijection-specific (selected by empirical input) — rather than as targets for substrate-Lagrangian closure — is the corrected research direction. The reclassification is internal to the framework, but the methodological pattern it instances is general: when a foundational program finds itself repeatedly stuck on a construction, the first check should be whether the construction is well-typed at the level it is being asked of.

24. THE ARCHITECTURAL PATTERN: PART III EXTENDED

Status: methodological comparison. Part III states separate no-go premise checks. Sections 22 and 23 add distinct gauge-invariance and definability questions about the same substratum and observable descriptions.

Part III distinguishes four constraints. Bell concerns local response factorizability and the setting ensemble; Kochen–Specker concerns valuations of a noncommuting observable algebra; Pusey–Barrett–Rudolph concerns preparation independence of ontic composites; Frauchiger–Renner concerns the joint use of observer-relative predictions. Describing each in terms of a substratum, an observable law and a trace-out identifies the relevant objects, but does not imply

that its premise fails or that its conclusion is evaded.

The §22 gauge-invariance no-go and the §23 category-error finding concern different restrictions. A substrate-only computation that is invariant under the established observation-preserving transformations cannot distinguish representatives within that equivalence class. A substrate-level Lagrangian requires an object with the differentiable structure used in its definition; the finite bijection alone does not supply it. Each argument has its own hypotheses and conclusion, rather than following from the failure of a quantum-foundations premise.

The comparison is therefore between the objects and premises of six arguments, not a common mechanism proving six conclusions. The fixed-basis correspondence, the Bell ceiling and the operational-completion characterization constrain different parts of that comparison. In particular, the trace-out does not establish H-Bell, choose a noncommuting operational algebra or determine the preparation-independence and observer-consistency assumptions.

A further no-go theorem requires its own mapping to the proposed completion and its own premise check. If that completion satisfies the theorem's hypotheses, it inherits the theorem's conclusion. Placing the theorem in the substratum or observable description is insufficient to avoid it.

25. THE OPERATIONAL AUDITS

Status: the audits themselves are developed in the framework's technical record; their methodological reading as systematic application of §22 to the framework's own content is the present part's synthesis.

The §22 gauge-invariance no-go is methodologically empty if not applied. A framework that proves a no-go theorem about substrate-only computation and then continues to claim quantities are substrate-derivable when they are in fact bijection-specific would have stated a result without using it. The framework's technical record applies the §22 result to its own content through systematic audits, which is part of what the operational methodology of this part means by "operational."

The audits proceed by going through each prediction the framework makes and asking, for each: is this quantity an equivalence-class invariant or bijection-specific? Where it is equivalence-class invariant, the framework's claim that it is substrate-derived stands. Where it is bijection-specific, the claim is corrected to reflect that the quantity is reproduced given empirical input rather than derived from substrate computation alone. The audits are not deflationary in net effect — the framework's substrate-derivable content is substantial after the audit is complete — but they are corrective on specific items, and the corrections matter for accurate characterization of what the framework claims.

Two audits in particular illustrate the discipline. A physics-sector audit covering the technical papers on the Standard Model and gravitational sectors classified twenty-nine predictions, finding approximately fifteen

substrate-derivable (i.e., equivalence-class invariant), four substrate-derivable in magnitude with bijection-specific directional content, and ten emergent-level mechanism-distinctive items that the framework's first-pass framing had not separated from the substrate-derivable cluster. A cross-domain audit covering chemistry, biology, and computational complexity classified thirty-four cross-domain claims under the same criteria; a mid-audit correction caught a generous-grading pattern and revised the substrate-derivable count downward, separating mechanism-distinctive items from substrate-derivable ones in line with §22's distinction. The numerical findings of both audits are in the framework's technical record; what is methodologically important here is that the audits ran on the same standard the framework's operational methodology proposes for foundational programs in general.

A third audit applied the same standard recursively to a previous expansion of the substrate-derivable count. The expansion had identified eight further candidates not covered by the prior audits and reclassified them as substrate-derivable; the recursive audit found that three of the eight survive a strict-criteria filter as approximately partially independent substrate-derivable items (with correlation discounts reflecting that they share machinery with already-counted items), four are mechanism-distinctive in §22's sense (distinctive in the framework's *explanation* for outcomes that the framework shares with the standard account), and one is framing rather than predictive. The recursive correction is methodologically the cleanest of the three findings: a framework that catches its own corrective moves inflating, and walks the inflation back to what survives strict criteria, is exhibiting the discipline the operational methodology recommends to others.

What the audits demonstrate as methodology is the second-order claim of this part. The §22 gauge-invariance no-go states that substrate-only computation cannot determine bijection-specific quantities; the audits use this no-go to classify the framework's own predictions; the resulting taxonomy distinguishes substrate-derivable content (equivalence-class invariants) from bijection-specific content (selected by empirical input) and from mechanism-distinctive content (distinctive in explanation though not in prediction-as-such). The taxonomy is what the framework offers in place of an undifferentiated count of "predictions." It is also what the operational methodology proposes for any foundational program of the same shape: not a count, but a categorized inventory in which each item is labeled by its relationship to the framework's gauge group.

The audits' general lesson concerns what the methodology asks of its users. A framework that proposes physics modulo gauge as its methodology takes on an obligation to apply the methodology to its own claims, and to revise those claims where the operational classification requires it. The framework's technical record shows that obligation being met more often than not — including in the recursive case above, where the obligation extended to revising the framework's own audits — though not always in the first pass. The methodological refinements that this discipline canonizes are the subject of §26.

26. THE OPERATIONAL REFINEMENTS

Status: the refinements stated here are derived from the framework's audit experience and are offered at the generality the operational methodology requires. The discipline they specify is canonical within the framework's technical record at multiple-instance strength; whether it generalizes to other foundational programs is the proposal-status claim.

The audit experience canonizes three methodological refinements that the operational methodology recommends as standards for any foundational program of the kind Parts II–IV describe. Each is stated below at the generality the methodology requires, with the framework-specific instance noted for reference but not relied on for the generality of the claim.

First refinement: verify before recording. When a finding adjusts a framework's count of substrate-derivable content — whether upward (more items than the prior count suggested) or downward (fewer items than the prior count suggested) — the finding should be subjected to strict-criteria filtering before being recorded as the new count. Positive findings inflate at first pass: the natural pattern is to count items that look as if they should qualify, and to discover on closer inspection that several of them do not meet the strict criteria. The discipline of running the strict-criteria filter on every count-change finding cuts the inflation, often by a non-trivial fraction. The framework's audit record contains eight instances of this pattern across one session of work, at strengths sufficient to canonize the refinement at the framework's internal level. The general claim is that any foundational program whose methodology proposes a categorized inventory will face the same inflation pressure, and the same strict-criteria filtering is the corrective.

Second refinement: recursive scrutiny. A corrective finding — one that catches an inflation in a prior finding — is itself subject to the same inflation pressure. The framework's audit record contains one instance of this case (a corrective finding that itself inflated the framework's count of substrate-derivable content by a factor of three at first pass, and that walked back to a strict-criteria count after a second-order audit). The refinement is provisional in the framework's own canonization scheme — second-order corrections are rare enough that one instance does not establish the pattern at the same strength as the first refinement — but the methodological point is general: self-correction does not stop at one pass, and the strict-criteria discipline applies recursively. A framework that catches the inflation in its own corrections is exhibiting the methodology more sharply than a framework that catches only the inflation in its original findings.

Third refinement: the observable/derivable distinction. A foundational program's open-question list is not a uniform category; it contains questions that fall into at least four distinct buckets. Some questions concern *observable* content the framework does not yet predict — open empirical questions in the standard sense. Some concern content that is *derivable* from the framework's

existing structure but has not yet been derived, requiring further calculation rather than further structure. Some concern content that is *structurally permitted* (consistent with the framework's commitments) but whose specification is gauge — equivalent representatives produce different specifications, none of which has empirical content. Some concern content that is *structurally forbidden* — gauge-incompatible specifications, ruled out at the level of the framework's structure. The four buckets ask different things of further work, and mixing them in a single "open questions" list obscures what each kind of question requires for closure. The methodological refinement is to distinguish the four buckets explicitly. The framework's open-question list, when classified by this distinction, is shorter than the unclassified list suggests, because some apparent open questions resolve to "structurally permitted but gauge" — not closures, but acknowledgments that the question has no answer-bearing content.

The three refinements share a structure. Each takes a category that a foundational program's working vocabulary collapses — the category of "predictions," the category of "self-correction," the category of "open questions" — and disaggregates it into sub-categories that ask different things and admit different kinds of further work. The methodology's discipline, throughout Parts II through V, has been the disaggregation move: track what is committed, distinguish it from what is not, and at each level find that the distinction the puzzle was failing to track is what makes the puzzle tractable. The refinements of §26 are that move applied to the methodology itself: track what kind of operational claim is being made, distinguish derivation from inflation from declension, and the methodology gains the same precision it asks the framework's content to display.

27. CONCLUSION

This paper has articulated the methodology of the Incompleteness framework through five commitments, and it is worth stating, in closing, what holds them together.

Part I distilled the framework's foundation and found it was two axioms rather than one, related by a no-go: the foundation stated at the exact granularity of what it commits to, with the single non-evidential posit named and isolated. Part II found that the framework's structural realism is constructive rather than programmatic **for the local reconstructed residue**: that residue is exhibited and shown unique under the stated conditions, and its relevant gauge freedom is exhibited, with exhaustive classification open; Bell-inclusive completion uniqueness remains outside the theorem. Part III organized the no-go theorems' distinct premise checks using the common substratum, reduced description and trace-out architecture, retaining Bell's H-Bell obligation and each case's separately stated assumptions. Part IV drew these together into physics modulo gauge: the proposal that the well-posed question for fundamental physics is the equivalence class and its representatives, not a single crowned fundamental. Section 19, within Part IV, identified that gauge freedom is one species of the framework's own foundational thesis — observational incompleteness — so that

the methodology and the physics are, on that species, one principle, and the methodology is revealed as the general form physics takes for an observer inside the system it describes. Part V developed the operational form of the methodology: two further no-go results (gauge-invariance of substrate-only computation; the category mismatch of a substrate-level Lagrangian), their comparison with Part III's distinct premise checks within the same architectural vocabulary, the systematic audits that apply the operational standard to the framework's own content, and three methodological refinements canonized by the audit experience.

The unifying thread is a single discipline: track precisely the distinction between what is committed and what is not. At the foundational layer this is the distinction between the indubitable axiom and the separable posit. At the level of structure it is the distinction between the equivalence class and the classified gauge freedom. At the level of the no-go theorems of quantum foundations it is the distinction between each theorem's premises and the substratum or operational objects they constrain. At the operational level it is the distinction between equivalence-class invariants (substrate-derivable) and bijection-specific quantities (selected by empirical input), and between either of these and constructions that are not well-typed at the level they are being asked of. And as a general methodology it is the distinction between the well-posed questions physics can answer by theorem and the ill-posed question — which representative is real — that it cannot. In each case the methodological move is the same: find the distinction the puzzle was failing to track, make it precise, and the puzzle either resolves, is shown to have been ill-formed, or reduces to a stated premise check.

That discipline is replicable, which is the paper's reason for stating it separately from the framework's physics. The framework's quantitative claims — the Standard Model retrodictions, conditional on the cubic-lattice commitment and the bridge hypotheses [SM] names (H-link among them, [SM §4.5]), the entropy coefficient, the neutrino prediction — stand or fall on their own evidence, in the technical papers. The methodology stands or falls on whether the discipline just described is sound and general. The two assessments are independent, and a reader may find the methodology worth adopting whatever they conclude about the physics, or worth rejecting whatever they conclude about the physics. We have marked, throughout, where the paper's claims are developmental: the foundational distillation of Part I is recent and offered for scrutiny; the structural theorems of Part II are summarized from the technical papers at the fidelity of faithful synthesis; the four no-go cases in Part III are developed to the scope stated there, two of them in this paper rather than in the technical papers and flagged for the referee accordingly; the generality of the Part IV methodology is offered as a proposal, not a result; the identification of gauge freedom with the gauge-incompleteness species of observational incompleteness, in Section 19, is offered explicitly as a proposal for scrutiny; and Part V is more recent still — its two no-go results are developed in the framework's technical record but their comparison with Part III's distinct premise checks is offered as a synthesis, the

systematic audits are themselves developed but the discipline they instantiate is proposed for use beyond the framework, and the three operational refinements are canonical within the framework's record at multiple-instance strength while their generality to other foundational programs is the proposal-status claim. What is not developmental is the discipline itself — track what is committed, distinguish it from what is not — and the claim that, applied to the foundations of physics, that discipline is what the framework's methodology consists in.

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This paper articulates, for a philosophy-of-physics readership, the methodological posture developed across the Incompleteness framework's technical papers. Part I develops the framework's foundational commitment; Parts II and III draw on structural results from the framework's substratum and main papers.

AI assistance in preparing this work is disclosed at
https://github.com/amaybaum/incompleteness#ai-assistance.

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